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BEYOND THE SQUARE PULSE: WAVEFORM SHAPE, EEG CORRELATES, AND THE PURSUIT OF NATURAL SENSATION IN TENS

by

JASON WHITSON

A thesis submitted in partial fulfillment of the requirements

for the Honors Undergraduate Thesis program

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ABSTRACT

This study investigates the relationship between electrical stimulus characteristics of shape and charge on evoked sensations and electroencephalogram (EEG) data during multi-waveform transcutaneous electrical nerve stimulation (TENS) of the median nerve. Neuromodulation methods have traditionally had little control over the location and quality of their associated evoked sensation (e.g., electric, vibration, touch). This experiment utilized five unique stimulus waveforms during TENS stimulation. EEG data were collected concurrently to provide an introductory objective measure of the neural responses underlying these sensory changes. Eleven participants completed three tasks (thresholding, super-threshold stimulation, two-alternative forced choice) using a two-electrode TENS approach. Stimulus waveforms were able to evoke sensations at notably different threshold charges and evoke significantly different intensities of sensation across notably different areas of the hand. Charge was found to have a complementary relationship with the evoked sensation's area and intensity across all waveforms. At threshold, the sine waveform evoked the most intense sensation over the largest area out of the five waveforms. Waveforms presented unique rates of recruitment between sensation categories, although unnatural pulse and electric sensations presented the highest rates across all waveforms. EEG data revealed significant differences in SEP amplitudes between waveforms at N140, P200, and P300, suggesting that the specific waveform shapes of sine and square, alongside charge, correlate with a higher amplitude of neural activity when compared to the triangular waveforms. A two-alternative forced-choice task showed that participants could accurately distinguish between waveforms (83.5%), even while reporting large differences in their subjective experiences. The findings highlight the importance of the temporal characteristics of stimulus waveforms, as most participants (90%) preferred non-square waveforms. The subjective findings suggest that

triangular waveforms offer optimal temporal parameters for evoking preferred sensations closer to touch, whereas square and sine waveforms consistently evoke intense sensations over larger areas of the hand.

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Throughout my entire life, I have received unwavering support from my mother, father, brother, grandmother, and grandfather. Without their support, I would never have had the confidence to become the individual I am today. Thank you for teaching me that I can do anything I set my mind to and to appreciate the simple joys in life. No matter the direction I take in my professional endeavors, I have never once worried that I would ever lose your support.

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To my friends, thank you for sticking by me through all the ups and downs of the last four years. It has been a joy to grow alongside each of you. To Sydney, thank you for supporting me through all the late nights, the studying, and the stress. You have made the most impactful years of my academic career possible. As this new chapter of life begins, I am grateful to have you by my side.

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INTRODUCTION

Approximately 185,000 people experience limb amputations in the USA each year [1]; a life-changing event that affects every facet of daily life. As a result of amputation, hundreds of thousands, if not millions, of sensory neurons are immediately lost. These sensory neurons are what make complex tasks, such as cracking an egg or turning a corner, feel like second nature. Without this sensory feedback, these tasks become much more difficult at best and near-impossible at worst. The importance of these sensations has been demonstrated in literature, as sensory restoration has been shown to improve gait in participants with transtibial amputation dramatically [2] and to substantially increase the success rate in a grasp task for participants with upper limb amputation [3]. In these studies, both invasive and non-invasive methods of administering stimulation are investigated. Although invasive methods offer superior localization, non-invasive methods provide an affordable, accessible means of administering electrical stimulation.

In this study, hardware and software methods are explored to administer transcutaneous electrical nerve stimulation (TENS) to the median nerve distally near the wrist to evoke sensory percepts in the participant's hand. Traditional TENS administration will be tested using various non-rectangular waveforms. During this experiment, EEG data will be collected and investigated to determine its relationship with the temporal qualities of a waveform, as well as with the quality and location of an evoked sensation. This study aims to examine the relationship between stimulus shape and charge on subjective experience of evoked sensations, and draw introductory conclusions on how these factors affect objective EEG data.

BACKGROUND

Basis for Sensory Feedback Study

Sensory feedback has long been studied to understand the inner workings of locomotion. Studies have shown that accurate movement requires information from a host of sensory systems [4]. This information assists in the predictive methods utilized during movement. Although these sensory systems exhibit some robustness, removing a sense directly impairs the animal's ability to complete a task [5]. The importance of sensory feedback is so paramount to locomotion that even robotic dog studies have used bio-inspired systems to improve the robustness of their movement [6].

Pivoting away from locomotion, upper limb tasks are similarly heavily reliant on our ability to obtain sensory feedback. In cases of Parkinson's and writer's cramp, this reliance is evident. A simple motor task of catching a falling weight demonstrated inconsistent grip force when tactile sensory feedback was inhibited [7]. In some cases, this inconsistency can be considered manageable; however, it can result in the destruction of a soft or fragile item. Individuals who rely on upper limb prosthetic devices often struggle with general coordination, such as grip tasks, for the same reason: they are unable to perceive essential characteristics of the item they are trying to grip. The reintroduction of artificial sensory feedback has shown direct improvement in coordination while operating an upper limb prosthetic, highlighting its importance [8].

Electrical Stimulation Methods

Through the exploration of sensory restoration, numerous stimulation methods have been discovered. Throughout all of these, the main distinction between any two methods is whether they are invasive or non-invasive. Invasive stimulation methods require a surgical procedure to introduce a stimulation device near or around the desired stimulation target. In contrast, non-

invasive methods are applied through the participant's skin and towards a desired stimulation target. This section will provide the necessary knowledge of both classes of stimulation for the presented project. Regardless of the method, the central ideas and definitions remain the same.

The rationale for the application of electrical stimulation as a whole is rooted in the Gate Control Theory of Pain [9]. This theory states that sensory input is modulated in the dorsal horn of the spinal cord by a gate mechanism in which large-diameter, myelinated Aβ fibers (typically responsive to touch and pressure) inhibit the transmission of nociceptive input carried by small-diameter Aδ and C fibers via inhibitory interneurons [9]. For example, when transcutaneous electrical nerve stimulation (TENS) is applied, it can activate Aβ fibers and generate activity trains along tactile pathways to the somatosensory cortex [10]. In cases of amputation, the cortical representation of the missing limb remains intact, allowing these activity trains to be interpreted by the brain as occurring on the phantom limb, thus creating an artificial sensation at that location. The activities elicited along the target tactile pathways vary greatly depending on the chosen stimulation method, waveform, and its parameters, allowing the type of sensation to be varied in intensity, location, and tactile experience (e.g., vibrating, pulsing). However, the direct relationship between stimulation and a specific evoked sensation remains unknown, leaving a significant gap in understanding that must be filled.

Typically, a stimulation waveform is described by the following metrics: frequency, amplitude, duration, period, duty cycle, and phase delay. Frequency dictates how many times per second a waveform will occur, whereas amplitude defines the magnitude (typically measured in milli-amps) of that waveform. Duration refers to how long a waveform will be administered and, in some cases, will be followed by the total number of output waveforms if the chosen waveform is not periodic. The stimulation period is the time required to administer one full wave of the

selected waveform, determined by the desired frequency and the "off" time between sequential waveforms. Duty cycle refers to the percentage of a period during which stimulation is on vs. off. For example, a 50 percent duty cycle with a 1-second period would have 500 ms of the stimulation waveform and 500 ms of no output. Finally, phase delay is the time between the anodic and cathodic phases (or vice versa). This delay is typically used when a non-periodic waveform is used, as a (generally square) cathodic or anodic pulse is sent after the initial waveform to offset its total charge. This is done to avoid tissue damage, as the overall charge of stimulation must be zero to avoid any long-term damage. For this reason, square waves have traditionally been used as stimulation waveforms. However, non-rectangular waveforms have begun to pique researchers' interest, as they present unique temporal information.

A direct relationship is created between the chosen parameters of a stimulation waveform and its associated charge. Based on the amplitude, frequency, duty cycle, and duration of a waveform, an overall charge of the waveform can be calculated for its associated anodic and cathodic periods. It has been observed that charge alone can have a leading effect on the evoked sensation that occurs, specifically increasing its perceived area and intensity [11, 12]. This relationship has been studied, but a comprehensive understanding has not yet been achieved.

Pulse trains are also a popular method of administering a stimulation waveform. A pulse train is a pattern of X waveform pulses delivered at Y frequency, repeated Z times, with a set delay between trains (this delay can be expressed as a percentage of the overall period, thereby defining the associated duty cycle of the period, not to be confused with the duty cycle of the waveform). The most notable of which is the Russian current, as shown here [13].

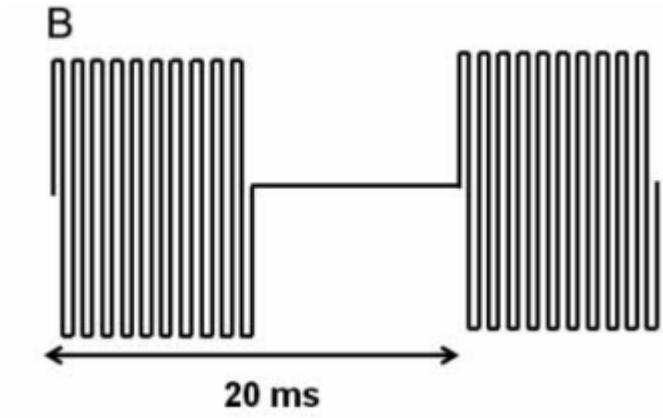


Figure 1: Example of Russian Current [13]

Invasive Stimulation Methods

Invasive stimulation methods have long been studied for targeting internal regions directly. Broadly, invasive stimulation provides superior localization of stimulation administration than non-invasive methods. This is due to its ability to circumvent the shape and amplitude changes a stimulation waveform experiences on its path from outside the body to the desired stimulation area. It can target a variety of areas, from large regions such as spinal vertebrae via electrode leads placed parallel to the spinal cord, to a single nerve bundle via peripheral nerve interfaces (PNIs). Importantly, invasive stimulation typically causes less discomfort at the stimulation site than non-invasive methods do, as they are able to recruit neural activity with a lower amplitude of stimulation.

PNIs are an invasive stimulation method that targets a specific peripheral nerve. PNIs have applications not only for active sensory restoration through stimulation but also for the passive recording and monitoring of nerves [14]. PNIs are specifically designed to modulate neural activity in a specific nerve or nerve cluster, enabling extremely accurate targeting. This allows researchers to investigate the contribution of particular nerves to specific movement or sensation areas.

In many studies on sensory restoration for the lower limbs, spinal cord stimulation is used. Traditional methods rely on implanted leads with electrodes extending their length, allowing stimulation to be administered at different levels of the vertebrae. These different vertebrae correlate to different limbs and locations on those limbs, providing a method to roughly target the area where the evoked sensation will occur.

Non-Invasive Stimulation Methods

Noninvasive stimulation techniques represent a promising area of investigation, owing to their relative ease of application, elimination of surgical requirements, and overall cost-effectiveness. Noninvasive stimulation typically causes discomfort at the electrode location and unintended stimulation of regions beyond the stimulation site. Further, distortion of the stimulus waveform between the administration side and the targeted nerve is a known issue that misrepresents the original characteristics of the stimulus when it reaches the target nerve. This limits the ability of researchers to draw robust conclusions between stimulus characteristics and their evoked response across participants, as the distortion differs based on the physical characteristics presented by each participant. Although these limitations persist, non-invasive stimulation has shown applications in various areas and offers significant potential for future improvements.

A leading noninvasive stimulation method is transcutaneous electrical nerve stimulation (TENS). In this method, superficial electrodes are placed on a participant's skin, and stimulation is delivered to target a location (typically a peripheral nerve) below the electrode sites. Relevant literature has shown the application of TENS for pain reduction and sensation restoration [12] as well as for bidirectional prosthetic control [15]. A recent study investigated the use of non-rectangular waveforms and how they elicit different types and locations of sensations during TENS

of the distal wrist [12]. This study found that non-rectangular waveforms could be viable alternatives to traditional square waves and that participants could distinguish different triangular waveforms. However, the findings on the driving factors behind how the waveforms felt different to the participants were inconclusive, partly because the participants were limited to using pre-selected adjectives to describe their sensations. Importantly, this paper lacked an objective method for measuring differences in sensation type between the non-rectangular waveforms, which could have provided insight into possible driving factors.

Temporal Interference Stimulation

Temporal interference stimulation is a novel, non-invasive stimulation method [16]. This method, studied extensively in deep brain stimulation, provides a technique for selectively activating deep targets without activating surrounding regions by using constructive interference. To do this, two waveforms are transmitted at slightly different frequencies from locations on opposite sides of a target area. The frequency difference between these two waveforms is called the "beat frequency" and should fall within the range of active neuronal activity. When these two waveforms interfere with each other, an envelope of that waveform is created that carries the beat frequency and an amplitude that is equal to the sum of both individual waveforms' amplitudes. Ideally, this interference will occur at the targeted location, activating the intended neurons without activating non-targeted locations outside of the interference envelope. This method is relatively new, and has various skeptics throughout the biomedical engineering field. The largest drawback that temporal interference presents is that it can place neurons between the targeted location and the electrode sites into a state of "conduction block", blocking neurons from firing.

EEG Data

Scalp electroencephalography (EEG) is the leading non-invasive method for recording and monitoring brain activity. This is done using electrodes placed on the scalp, which record microvolt-level voltage changes in the brain. These changes directly correlate to the summed extracellular potentials of millions of neurons [17], providing insight into the brain's reaction to different external stimuli. This ultra-low data collection magnitude is a major drawback for signal integrity, resulting in a very low signal-to-noise ratio. This low signal-to-noise ratio makes it difficult to draw conclusions from EEG data, necessitating a number of preprocessing stages before data analysis can occur. These preprocessing methods will be discussed further in the following section. Intuitively, an invasive method of monitoring brain activity, referred to as intracranial EEG, would certainly increase the signal-to-noise ratio of the data and thus improve its quality. However, invasive methods of brain monitoring not only require an expensive surgical procedure but also risk long-term biocompatibility issues between the brain matter and the monitoring device [18]. Due to these complications, many researchers use scalp EEG as their preferred method for recording brain activity, despite its inherent noise. The number and positioning of the EEG electrodes vary between devices, with the most common electrode setups containing 32, 64, or 128 individual electrodes. Electrode configurations for scalp EEG caps are referred to as montages and are typically standardized to the 10-20 or 10-10 electrode montages, which have been adopted by the International Federation of Societies for Electroencephalography and Clinical Neurophysiology [19]. These configurations allow specific brain regions to be targeted and compared, which is crucial for drawing conclusions about the relationship between external stimuli and activity in specific brain regions.

Due to the microscale at which EEG is recorded, the raw data it captures are highly susceptible to noise and external artifacts. Fortunately, the optimal pre-processing methods are well documented in the extensive research on this data. A common pre-processing pipeline for EEG data should consist of the following procedures. First, data is band-pass filtered using a 0.5 Hz high pass filter, and anywhere from a 40 Hz – 190 Hz low pass filter. These values are chosen to both remove slow drift artifacts through the 0.5 Hz high-pass filter and suppress muscle activity and other high-frequency noise through the low pass filter. A notch filter is then applied to remove power-line noise, which varies from 50 Hz to 60 Hz depending on the local power-line frequency. Typically, a re-referencing is then applied across all electrodes to provide a common scale between electrodes [20]. This re-referencing typically uses the average of the two mastoid electrodes, as they provide the most electrically neutral reference point due to their proximity to the bony area behind the ears. Finally, artifacts such as blinking or movements must be removed from the overall dataset. There are numerous methods to do this, with the most popular being independent component analysis (ICA) applied to the data [21]. This method utilizes the spatial representation of EEG data, which is dominated by the very artifacts researchers desire to remove. Components found to contain these artifacts are flagged and then removed from the data, resulting in a final artifact-free dataset. A final global re-referencing is typically applied to provide a comparable scale between electrodes, allowing comparison to draw meaningful conclusions.

After these preprocessing steps, a final bandpass filter can be applied to the preprocessed data. This filter is typically used to highlight certain brain activity frequencies for analysis, removing noise from non-target frequency bands. In general, brain activity can be divided into five

frequency bands, each representing a different type of activity. The delta (1 Hz - 4 Hz) and theta (4 Hz - 8 Hz) bands are associated with deep sleep and drowsiness, respectively, and are not typically considered for analysis during experiments concerning activities conducted while the participants are awake. The alpha (8 Hz - 13 Hz) and beta bands (13 Hz - 30 Hz) contain information that is extremely relevant to data processing and cognitive tasks. The alpha band includes relaxation and idle information, as well as motor cortex activity important for sensory experiments. This motor cortex information is sometimes separated from the alpha band into the "Mu" band, which is measured only over motor cortex electrodes. The beta band contains information that is the most correlated to active thinking, leading to its common application in EEG analysis pipelines. The combination of information from both the alpha and beta bands has been used to train EEG-based BCIs [22], highlighting the important features these bands exhibit. Above these bands lies the gamma band, which stretches from 30 Hz to 100 Hz. This band correlates to the somatosensory cortex, sensory processing, working memory, and general attention. However, the information presented by the gamma band is difficult to analyze due to its high frequencies. Thus, researchers must take direct considerations into account when analyzing this band to ensure that preprocessing steps enable meaningful conclusions.

Once band-pass filtered to the desired frequency band, or left unfiltered for broadband analysis, EEG data is visualized through averaging many epochs to produce general event-related potentials. An epoch consists of EEG data plotted over a defined interval before and after an external event. Typically, this event is an external stimulus, such as a color flashing on a screen or an audio signal being played [23]. In most experiments utilizing EEG data, this specific event would be repeated multiple times for each participant, creating multiple epochs. These epochs are

then averaged across all trials with the same stimulus, providing the average brain response to that stimulus. This is referred to as an event-related potential. When the event that occurs creates an evoked sensation, these event related potentials are sometimes called somatosensory evoked potentials (SEPs). To improve the correlation between this event-related potential and the stimulus, event-related potentials are typically averaged over precisely selected electrodes. For example, if the external stimulus was an audio signal, event-related potentials would be averaged over specific electrodes corresponding to the auditory cortex. These event-related potentials can be used to highlight important temporal information, such as local maxima or minima, which are typically referred to with a descriptor (P - positive polarity, N - negative polarity) followed by the time in milliseconds post-stimulus at which the extrema occurred [24] (such as P300, which is typically used for training certain BCI algorithms). In addition to visual analysis, event-related potential data can be analyzed using various statistical methods to highlight contrasts between stimulus classes.

Event-related potentials represent the leading method for visualizing and analyzing EEG data in the time domain. In addition to time-domain analysis, various frequency-domain analyses can reveal critical information that supports relevant conclusions in neuroscience. The power spectral density (PSD) is the squared magnitude of the FFT of an epoch, which converts the EEG data into the frequency domain while discarding temporal information. For a certain event, the PSD is calculated for each epoch and then averaged. This analysis technique provides insight into how much power each band possesses [25]. For example, an experiment collecting EEG data during sleep would expect most of its power to lie in the delta and theta bands. The leading method for denoising PSDs is Welch's method [26]. Welch’s method splits an epoch into overlapping

windows and computes the PSD for each window. These PSD windows are then averaged to produce a single PSD with low overall variance. However, this method sacrifices frequency resolution to achieve this lower variance. There are further frequency-domain analysis techniques that can preserve some time-domain features, namely wavelet transforms. However, these techniques are outside the scope of this thesis.

EEG data has also shown promise for use in machine learning, with extensive use in training BCI classification algorithms. These algorithms use the same process as previously detailed and extract spatiotemporal features from their training data (which consists of various event-related potentials). A common conclusion drawn from BCIs trained on multi-participant EEG datasets is that they exhibit high subject specificity. Due to large variations in EEG data across participants, BCIs are typically entirely subject-specific, meaning a classification algorithm must be trained for each participant independently of others to achieve the highest accuracy [27]. This raises a major drawback, creating a large training overhead between data collection and classifier fitting. It also highlights the large variation between subjects' EEG data, even when that data is collected under the same testing conditions. Creating machine learning classification algorithms that can maintain high accuracy across training data collected from multiple participants remains a pertinent problem that many researchers are actively working to solve.

Neuromodulation and its Effect on EEG Data - Current Works

Current literature on neuromodulation (specifically TENS as it pertains to this thesis) has shifted focus from pain management to investigating the relationship between different stimulation patterns and the influence of waveforms on EEG data [27]. EEG data has been widely used as a

data collection tool, as it provides objective measurements for studies that have historically relied solely on subjective self-assessments. The ability to map objective changes in EEG data enables researchers to optimize neuromodulation techniques, potentially enhancing sensory feedback for prosthesis users [28].

A critical finding of these research articles is the identification of reliable biomarkers in EEG data that can be used to identify brain events, such as sensory processing. These biomarkers are present in event-related potentials. The literature has reported that TENS-evoked sensations in both amputees and non-amputee subjects consistently exhibited N100 and P200 peaks, and has associated the N100 peak with early sensory processing and the P200 peak with the subject's perception of the stimuli [28]. Changing the TENS stimulus parameters has been shown to affect these peaks. Specifically, pulse-width-modulated and non-modulated high-frequency TENS have been studied and have resulted in reductions in the N100 peak amplitude and in the band power in the theta/alpha band [29]. This direct effect on these two ERP peaks provides a basis for future analyses and offers preliminary evidence that varied waveform parameters can produce different neuronal responses.

Further, recent literature has provided correlations between band power dynamics and various sensation qualities of the evoked sensation via TENS. It has been found that an increase in alpha band power (driven by changes in TENS parameters) correlates with a suppression of a subject's perception of a pain sensation's intensity [30]. Further studies have found a relationship between TENS and modulation of beta-band activity, suggesting that this band contains significant information regarding neuromodulation and its effects on the sensorimotor cortex [29]. These

findings shape future EEG studies by identifying important frequency bands to investigate, highlighting continuity with previous results and possible deviations from literature.

Neuromodulation and its Effect on EEG Data - Gaps in Knowledge

Despite the depth of research devoted to this topic, significant gaps in knowledge remain and require further investigation. It has been demonstrated in various studies that a reduction in sensation intensity correlates with a suppression of alpha-band power and a reduction in ERP peaks. However, it is currently unknown how different types of waveforms affect EEG data, or how their associated sensation qualities influence EEG data. The ability to distinguish sensation intensity from EEG data alone provides significant promise for distinguishing different sensation qualities as well. In the pursuit of addressing this gap, it is also unknown how inter-subject differences may affect EEG data's ability to discriminate between sensations, as the definition of a specific sensation may differ from subject to subject. Despite these challenges, the ability to distinguish the information about a stimuli and its evoked sensations from EEG data would represent a breakthrough in sensory restoration applications, enabling naturalistic sensations to be generated through a feedback loop that modifies stimulus parameters to achieve a desired EEG response.

To fully achieve sensory restoration, electrical stimulation must also be able to target certain locations. This represents the second major gap in knowledge, present in both pure TENS neuromodulation studies and TENS neuromodulation studies involving EEG. The relationship between stimulation parameters and the location of evoked sensation, as well as the relationship between a location of evoked sensation and EEG data, are both relatively unknown. The discovery

of a driving factor would enable targeted sensory restoration and solve a major problem prosthesis users face.

Further conclusions about the relationships among different stimulus parameters, charge, evoked sensation qualities, and evoked sensation locations could be drawn from EEG data. The literature has established relationships between sensory processing and perception at the N100 and P200 ERP peaks, and further relationships could be identified in future studies. A deeper analysis linking changes in stimulus parameters or subjective experience to their effects on the frequency content of EEG data could yield conclusions about the important driving factors behind a subject's experience of certain stimulation.

METHODS

Motivation

TENS has shown promise in sensory restoration in both neurologically intact and damaged individuals. TENS has typically been used with rectangular waveforms, as is common in electrical stimulation methods [12]. The main drawback of rectangular waveforms is the unnatural sensation they tend to elicit during stimulation. Non-rectangular waveforms could provide sensations that more closely approximate a natural experience; however, the current literature lacks objective data to support this claim. Collu et al. addresses this question by testing four non-rectangular waveforms and comparing their evoked sensation with sensations evoked by a traditional rectangular waveform. The non-rectangular waveforms consisted of sinusoidal, triangular, linearly increasing, and linearly decreasing. This study concluded that non-rectangular waveforms could be a viable alternative to traditional rectangular waveforms, as they elicited distinctly different

sensations across different locations of the hand; however, it relied on subjective data to define naturalness and to record how the evoked sensation was experienced. From the base provided by Dr. Collu, it can be seen that changing the waveform of TENS stimuli creates subjective differences in sensation, which are hypothesized to arise from biophysical responses that differ between waveforms.

From the data already collected from subjective measurements across participants, the need for a truly objective data category for comparison has become strikingly apparent. Similarly to the use of EMG data when recording biophysical responses to spinal cord stimulation, EEG data has been used in conjunction with TENS, however, much more sparsely. EEG data serves as the objective measurement that provides insight into possible drivers for a collected subjective dataset. For example, if one participant described a square waveform as *electric* and a sine waveform as a *touch*, the EEG data for these waveforms could be epoched and compared using various statistical methods to determine whether a significant factor influenced this decision. Extending this analogy, if the participant reported that certain trials with various waveforms felt like a *touch* only 25% of the time, EEG data could be analyzed to determine whether these touch-sensation trials shared a common denominator. Ultimately, conclusions drawn from objective EEG data could provide a method for shaping stimulus waveform parameters to elicit desired event-related potentials and steer the sensation experience.

For naturalistic sensation to be restored, it's important to emphasize the significance of how it feels, rather than simply whether it occurs. Traditionally, the "natural" characteristic of a sensation has been measured through subjective word selections, i.e., "How *natural* does this feel?" (Figure 2) These terms have different meanings for each participant and present difficulties for non-English speakers. The subjectivity of traditional analysis leaves room to explore

potentially objective methods that yield meaningful data, and drives this thesis to utilize EEG data as its objective data collection method.

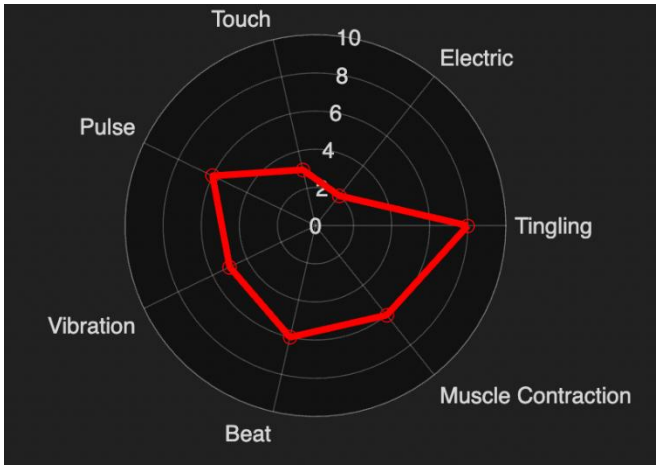


Figure 2: Example Adjectives Used to Describe a Sensation

Procedures

Hardware

Waveforms were generated through a Measurement Computing DAQ and converted to current through the use of an STMISOLA LINEAR ISOLATED STIMULATOR. These waveforms were confirmed to have the correct shape and length through an oscilloscope. To ensure that waveforms output as desired during the experiment, a CBLCFMA current splitter was attached to the STMISOLA LINEAR ISOLATED STIMULATOR'S negative output port and fed back into one of the DAQ's analog input channels. This study used a TMSi SAGA 64-channel EEG system to record EEG data. Electrodes will be placed corresponding to the median nerve of the participant. The median nerve of the participant will be located using a portable ultrasound device. Once detected, the median nerve will be marked on the participant using a skin-safe marker, and the ultrasound's image will be saved to ensure proper location of the median nerve post-experiment. For traditional TENS, a pair of electrodes will be placed on the participant. The anode electrode

will be placed approximately 1-2cm from the wrist joint, and the cathode electrode will be placed 3 cm proximally from the center of the anode. Both electrodes will be placed directly above the median nerve as marked by the experimenters.

Traditional TENS Waveforms

For traditional TENS administration, the same five waveforms used in Collu et al. [12] were selected. The waveforms consisted of a biphasic wave with balanced charge and an interphase delay of 100 μs, which is considered safe for long-term neurostimulation. The cathodic phase of the waveform consisted of 5 shapes: sine, square, linear increase, linear decrease, and triangle (Figure 3). Between these five, two subclasses can be referred to as non-triangular (sine and square) and triangular (linear increase, linear decrease, and triangle). This is followed by an anodic phase consisting of a square pulse that balances the overall charge in the waveform. For these waveforms, frequencies were chosen based on the waveform lengths investigated by Collu et al. Specifically, we have decided to use 400 μs for our cathodic phase, which will correspond to a frequency of 1250Hz (as we will just use the first phase of the biphasic waveform). This was chosen as a middle ground between the range used in the study mentioned above, which correspondingly provided a relatively low overall charge. The relationship between charge and elicited sensation will be explored in the study's second task, thereby making this original low-charge a vital baseline against which waveforms with higher charges can be compared. Waveforms will be sent as a single pulse to negate the dominance of charge in our findings during traditional administration.

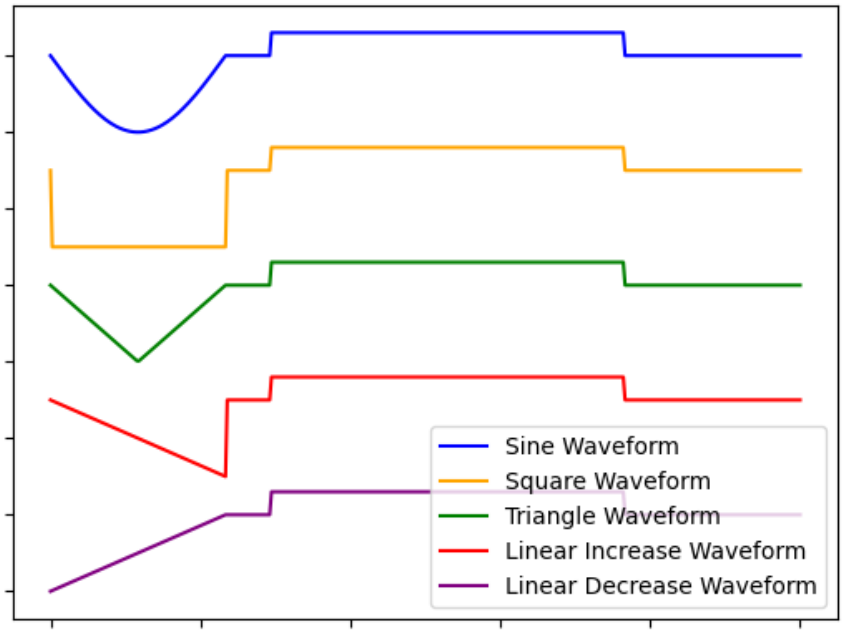


Figure 3: Example Waveforms

Experimental Procedures

EEG Data Collection and Subject Self-Reporting

At the start of each experiment, the participant will be fitted with a 64-channel Saga EEG cap. This EEG cap will record data throughout the entire experiment, with the recordings segmented between tasks. To ensure epochs can be extracted correctly, a unique trigger will be provided for each waveform-amplitude pair in each task. The use of these unique triggers also protects against error triggers, as preprocessing can detect and remove unexpected triggers for a given trial, ensuring that epochs use only relevant data points. The trigger will be sent at the start of each waveform for the three stimulation tasks. These rigid timings ensure epochs are averaged with the same onset between waveforms, tasks, and participants.

During the electrical stimulation tasks, the participant will experience evoked sensations. These evoked sensations have been shown in literature to vary in perceived location and quality between individuals, even when the stimuli are the same. To record these differences, the participant will be prompted to report the quality and location of the evoked sensation. These self-reported variables will be used in EEG data analysis to further separate epochs by specific sensations, drawing conclusions about their associated effects on EEG data.

Participants

Eleven healthy subjects (nine male, two female) completed the experimental procedures for the traditional administration of TENS. Two participants mistakenly received four amplitude levels rather than three for certain waveforms (sine and linear decrease, respectively) within task three, and thus their information was excluded from amplitude-specific analysis for those waveforms.

Task 1 - Threshold Calculation

Various participant physical attributes influence their biophysical response to the same stimulus. To reduce this variation, a base detection threshold for each participant will be determined before the bulk of data collection is performed. To determine these thresholds, the selected waveform was sent at 3-second intervals. The participant was facing a computer screen displaying a GUI that instructed them to interact when a sensation was detected. When not detected, the waveform amplitude would increase by 250 μA steps between administrations. When a sensation was detected, the participant would interact with the GUI to report that a sensation was elicited, and report its qualities as defined previously. To set the thresholds, a one-up, two-down adaptive psychophysical method was used, with a threshold of 5 reversals. These thresholds

establish the baseline minimum stimulation amplitude per waveform for the remainder of the testing session.

Task 2 - Super-Threshold Stimulation

Stimulation waveforms will be examined at and above threshold amplitude to discover relationships between waveform, amplitude, sensation location and quality, and associated EEG data. To collect this data, waveforms will be sent at the three separate stimulation amplitudes. For traditional TENS, amplitudes will be divided into low (at threshold), medium (threshold + 250 μA), and high (threshold + 500 μA). Each waveform will be sent 10 times at each amplitude level, totaling 150 trials of data collected per participant from this task. As the amplitude increases, the waveform's overall charge increases. The relationship between charge and perceived sensation has been studied in prior literature, adding another variable to the list of factors influencing elicited sensation. This data will be used to explore this variable and its relation to perceived sensation.

Task 3 - Two Alternative Forced Choice

Although different waveforms have distinct parameters that influence biophysical responses, it remains unknown whether the participant can distinguish one stimulus from another when blinded. To answer this question, waveform types will be paired and administered to the participant sequentially. Rather than having the participant distinguish these waveforms by selecting which waveform shape the stimulus was, which may be difficult to understand or derive results from personal bias, the participant will be asked to choose which of the two waveforms feels more "natural". Although this question itself rests on a subjective measure of naturalness, it provides insight into whether waveform parameters can influence the quality of sensation to the extent that the participant can consciously distinguish one waveform from another. The order of

the waveforms within the pair will be randomized between administrations to ensure that participants detect the new stimulus consistently rather than following a pattern. All ten possible combinations of the five waveform bases will be utilized during this task. The waveforms will be delivered at their threshold amplitudes to minimize the effects of suprathreshold stimulation on perceived sensation. After the two waveforms have been administered, the participant will simply report which of the two waveforms felt more "natural". Once the participant has selected one waveform as the more natural out of the two ten times, the next set of waveforms will be sent.

EEG Data Pre-Processing

EEG data was collected through the use of a 64-channel SAGA TMSI EEG headcap system, which was set to a sampling rate of 1000 Hz. Before and after the experiment, all electrodes were confirmed to have less than a 5 kΩ impedance. The general pre-processing steps used in this experiment have been adapted from Alibou et al. to allow comparisons to be drawn between the datasets [34]. The data was re-referenced to the two behind-the-ear channels of M1 and M2, due to their close proximity to the participant's bone structure. A 0.5 Hz, 94th-order Chebyshev Type II filter was applied to remove slow drift artifacts, followed by a 60 Hz comb notch filter to remove power line interference. Bad channels were automatically removed based on preset rejection criteria, and then all electrodes were re-referenced to the global average. An ICA was then fit to the data to remove non-neural components that EEG data may capture, such as blinks and muscle movements. Artifacts presented from the stimulation were removed through blanking samples around the stimulation event (-2 ms pre - 10 ms post), and a 190 Hz low-pass filter. Baseline correction was used for each epoch to be used for SEP analysis to ensure data was compared using similar scales that are invariant to the general magnitude of brain activity not related to the stimulus

at that time. SEPs were averaged over electrodes found to be relevant to the event, such as those that correlate to the sensorimotor cortex. Specific regions of interest will be identified based on the number of significant events.

RESULTS

Evoked sensation and sensation location

The various stimulation waveforms were detected by participants at notably different amplitudes. These amplitudes corresponded to different overall charges due to the temporal differences presented by each waveform. Thus, each waveform should be considered both in terms of overall amplitude (mA) and in terms of overall charge (μC). A description of how these different shapes affect charge amplitude is summarized in Table 1.

Waveform	Charge Per 0.25mA Step
Square	0.1000 μC
Sine	0.0637 μC
Triangle	0.0500 μC
Linear Increase	0.0500 μC
Linear Decrease	0.0500 μC

Table 1: Charge differences presented by amplitude steps across waveforms. These differences are due to the temporal characteristics of the waveforms

The square waveform had a significantly lower threshold amplitude than the other four waveforms, which correlated to the highest overall charge (significantly higher than the triangle waveform only, $p = 0.039$). The sine waveform had the 2nd lowest threshold in amplitude, yet the 2nd highest threshold when measured in charge. The three triangular waveforms resulted in similar thresholds and, thus, similar charges. The charges at which the three triangular amplitudes were detected are, interestingly, around one amplitude step (respectively) lower than those at which the square or sine waveforms could be detected, highlighting that triangular waveforms may provide a superior ability to elicit "softer" sensations due to the lower charge required to evoke a sensation that participants perceived.

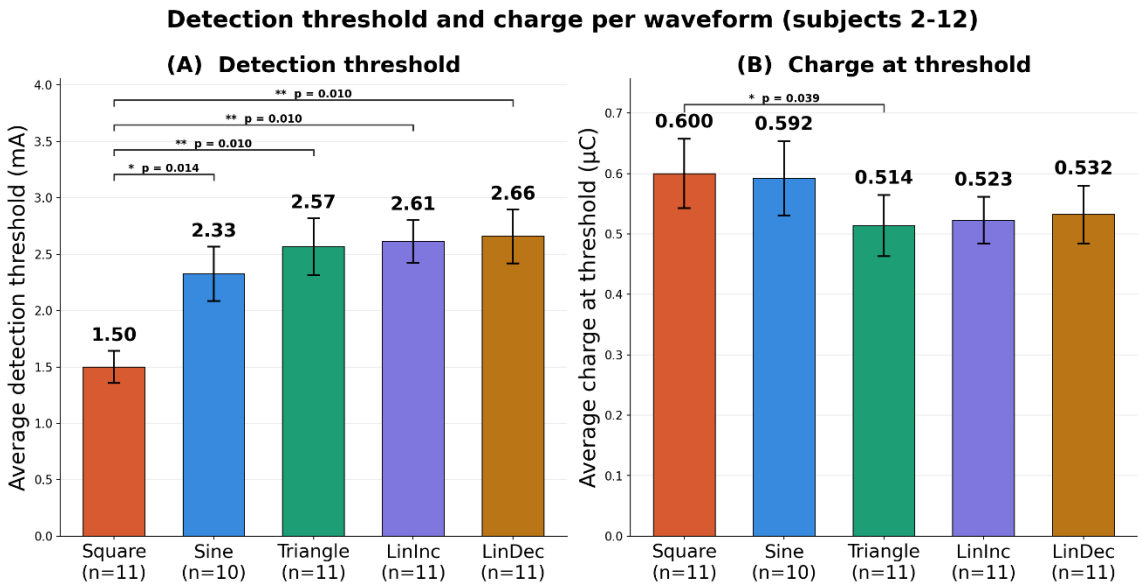


Figure 4: Threshold amplitude and associated charge revealed that sine and square both required the highest charge to evoke a sensation. The three triangular waveforms evoked sensations at a charge step approximately one step lower than the non-triangular waveforms, suggesting they may evoke less intense sensations.

As expected from previous literature, different stimulus waveform shapes correlated to changes in sensation location across the three amplitudes. The sine waveform yielded the highest overall area of evoked sensation across participants, with an average of 4.2% of the total hand area experiencing an evoked sensation each time the waveform was administered. This was followed by the square waveform at 3.7%, the linear decrease waveform at 2.9%, and the linear increase and triangular waveforms tied at 2.8%. Interestingly, none of the waveforms evoked sensations on the back of the hand, resulting in relatively low percentages, as 50% of the hand is completely unstimulated, no matter the stimulus or intensity. This is somewhat expected, as most of the back of the hand corresponds to the radial and ulnar nerves; it is unexpected, however, that across all trials, there was such limited off-target stimulation. These results highlight the influence of the waveform's shape, as the sine waveform, which had a lower average charge (0.044 μC lower) than the square waveform, was able to evoke sensation over a larger area of the hand. This suggests that specific temporal information contained by a stimulus waveform can be used to modulate the overall area of an evoked sensation, rather than relying on charge alone. The locations of these evoked sensations for each waveform are illustrated in Figure 5.

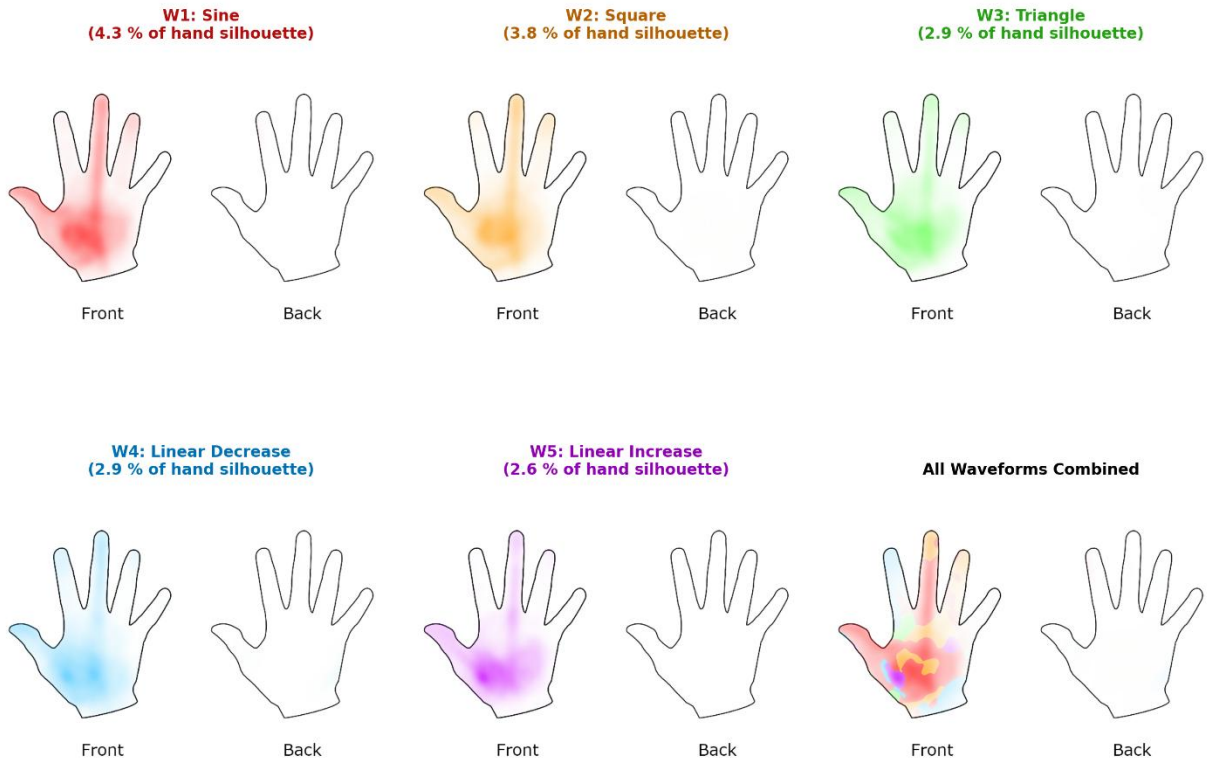


Figure 5: Grand average hand heatmaps for each waveform. The sine and square waveforms were able to evoke the largest average area of sensation (4.2% and 3.7%, respectively), while the three triangular waveforms evoked a much smaller area (2.8%, 2.9%, and 2.8%). This difference is not statistically significant due to the high inter-participant variation of the evoked sensation area.

From this visualization, it is evident that, regardless of stimulation parameters, the bottom-left section of the palm is the most commonly reported location for an evoked sensation. Different stimulation waveforms were able to evoke sensations in notably different locations of the hand outside of this area; the sine waveform commonly resulted in evoked sensations on the thumb, while the square waveform was able to evoke sensation on the tip of the middle finger. Interestingly, the three triangular waveforms did not indicate consistent sensation locations beyond the commonly observed palm area in this study.

The different stimulation waveforms further resulted in significant variations in subjective experience, suggesting that participants were able to (broadly) distinguish one stimulus from the other (this claim will be built upon during task four). The sine waveform presented the highest overall "pulse" sensation at a mean value of 1.48, significantly higher than any of the other four waveforms ($p = 0.006$). The square waveform presented the highest overall "electric" sensation at a mean value of 1.036 ($p = 0.001$). Although both of these values are relatively small, they showcase that stimulation waveforms can present unique sensation profiles compared to others. On average, square was able to evoke the highest average sensation intensity of 0.57, while sine closely followed by evoking an average sensation intensity of 0.55. Interestingly, the average sensation intensity of the three triangular waveforms (linear increase = 0.39, linear decrease = 0.38, triangle = 0.36) were significantly lower than either of the square or sine averages (Figure 6). The connection between average perceived stimulation intensity and average waveform charge is almost one to one, with only the linear increase and linear decrease waveforms switching order. This could suggest that the overall charge, rather than temporal waveform characteristics, is the driving factor for the perceived intensity of an evoked sensation.

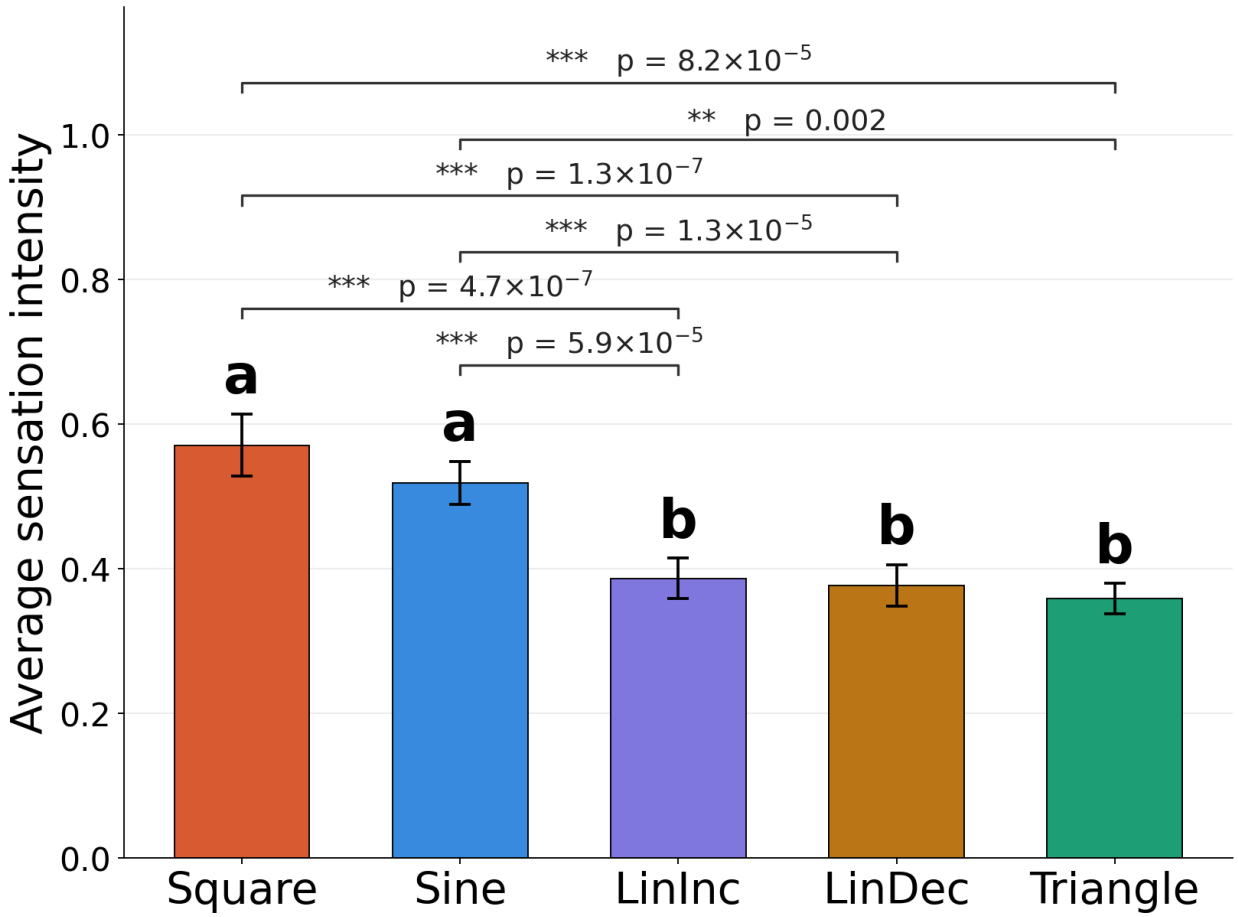


Figure 6: Average sensation intensity evoked by each waveform revealed significant differences between square and linear increase, linear decrease, and triangle, as well as between sine and linear increase, linear decrease, and triangle. Analysis highlights the two groups of stimuli formed through subjective experience, non-triangular and triangular.

Further, the two waveforms that present significantly high subjective average sensation values also present the two highest overall areas of evoked sensations. On average, the sine waveform evoked sensation across 4.2% of the participants hand, while the square waveform evoked sensation across 3.7%. Although these values are not significantly larger than the triangular waveforms, the relationship showcases a clear correlation between perceived intensity and evoked area. This relationship between sensation intensity and area suggests that their influencing factors, shape and charge, exert similar effects on both perceptual measures, rather than opposing effects (e.g., enlarging area while decreasing intensity).

The other 6 sensation types were scattered between the five waveforms, with no single waveform standing out from the rest. The contraction and needle sensations had the lowest means (0.23, 0.073), suggesting that future research could benefit from substituting these sensation variables with others that may be more directly correlated with sensations evoked by near threshold electrical stimulation. The sensation profile for each waveform is shown below.

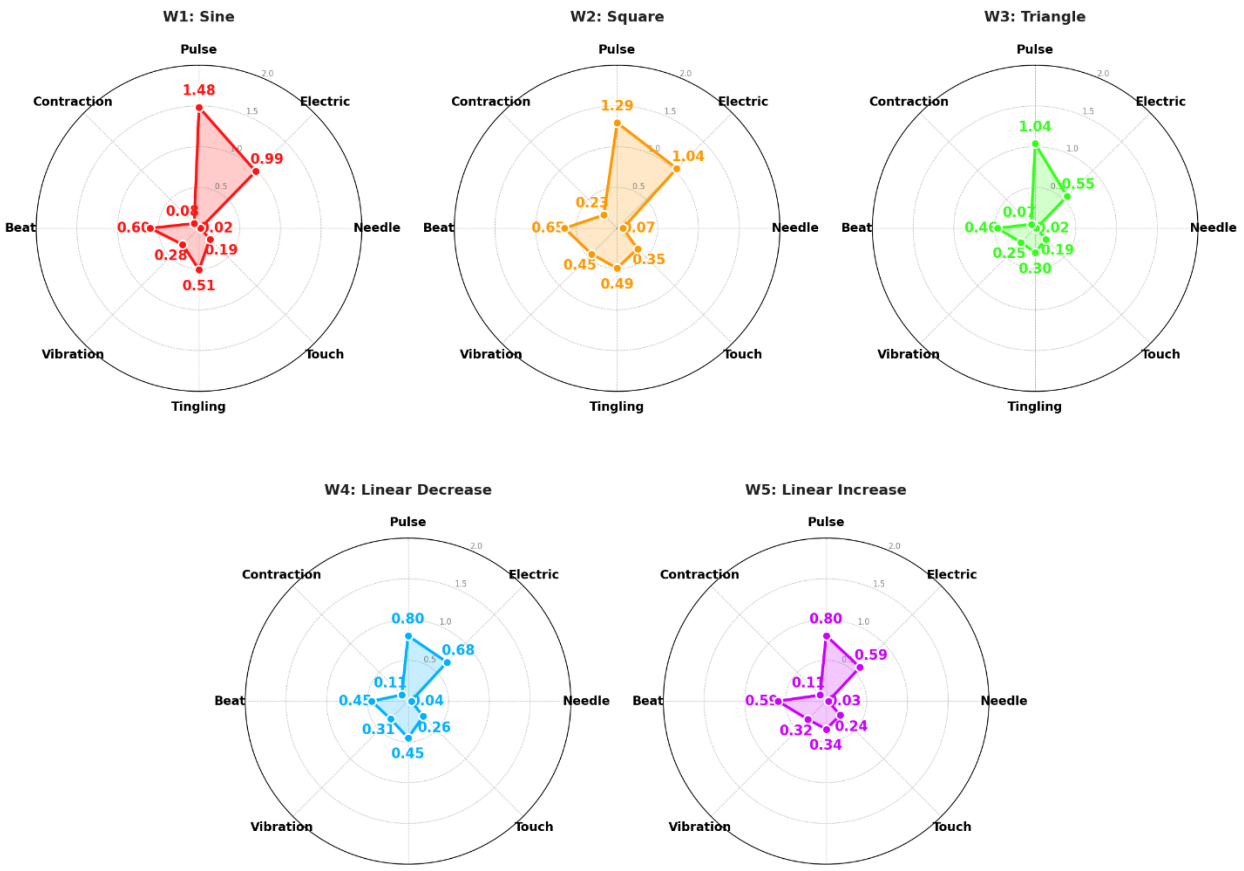


Figure 7: Average sensation per waveform. The intensity of the pulse sensation was significantly higher for the sine waveform (p =0.006), and the intensity of the electric sensation was significantly higher for the square waveform (p = 0.001). All other sensations were not experienced as significantly higher or lower between waveforms.

Interestingly, no waveform yielded a significantly higher mean value for the touch sensation, nor did any yield a significantly lower mean value. This is likely a result of the various drawbacks associated with non-invasive stimulation methods, such as the TENS used in this experiment. The notable differences in both sensation quality and location encourage future research to investigate this relationship among stimulus waveforms during invasive stimulation methods, which would provide superior localization of stimulation that would preserve charge and shape.

From these general sensation types, a secondary analysis emerged: does the type of sensation influence the qualities of the evoked sensation? Simply put, when a subject experiences an electric sensation or a touch sensation, do the qualities between those sensations differ on average? Before this question can be answered, we must understand the possible drivers behind these sensations. Between the eight sensations, the pulse sensation was associated with the highest average charge (0.65 μC), closely followed by vibration (0.6 μC). The difference between the charge associated with vibration and the next closest sensation, needle, (0.04 μC) is larger than the difference between needle and the sensation with the lowest average charge, touch (0.03 μC). This

showcases the large difference in charge between the two highest sensations compared to the other six. These charges are shown in the figure below.

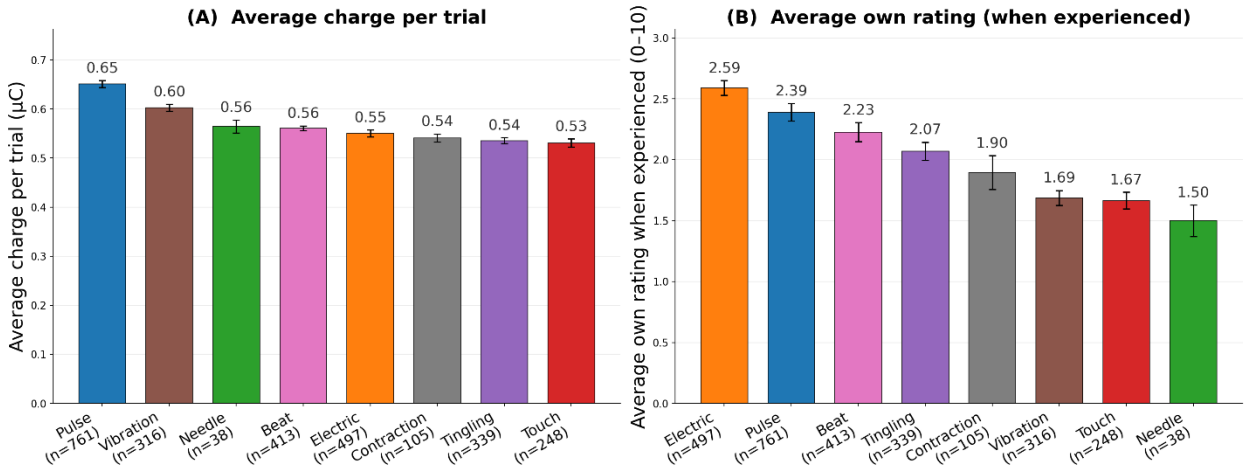


Figure 8: Charge and perceived intensity of unique sensations showed little relationship. The pulse and vibration sensations were evoked with a considerably higher charge than the other six sensations. The touch sensation was evoked with low charge and perceived with low intensity. The general asynchrony between charge and perceived intensity, when filtered by sensation type, is likely due to differences between subjective experience and personal definitions of these sensations.

Interestingly, the charge associated with the sensations does not clearly map to the sensations associated with perceived intensity. The electric sensation, which averaged 0.1 μC less charge than the pulse sensation, evoked the most intense perception. Vibration, which averaged the second-highest charge, was perceived as the third-least intense sensation. This general mismatch between charge and intensity within sensation classes highlights the impact that the specific waveform shape has on the perception of the sensation.

Specific waveforms exhibited higher correlation with evoking certain sensations compared to others. The pulse waveform was evoked by the sine waveform frequently, whereas the electric sensation was evoked by both the sine and square waveforms at a higher rate than any of the three

triangular waveforms. Interestingly, the beat sensation was evoked the most by the linear decrease waveform, at a rate much higher than the other waveforms. The touch sensation was also most commonly evoked by the linear decrease waveform, closely followed by the square waveform. These results showcase that waveform shape does not directly correspond to a singular sensation, likely due to physiological differences between participants and the associated distortion the waveform shape experiences while propagating to the intended nerve. However, it is clear that different shapes correspond to different rates of recruitment between different sensations. Although the rate of recruiting a specific desired sensation across any waveform peaks at 53% (sine recruiting pulse), the secondary analysis concerning the rate of not recruiting undesired sensations should play an equally important role when selecting the desired waveform shape. If the goal of a study was to evoke the touch sensation, the linear decrease waveform should be selected not only for its ability to evoke the touch sensation, but also for its ability to reduce the frequency at which the most unnatural sensations of pulse and electric are evoked.

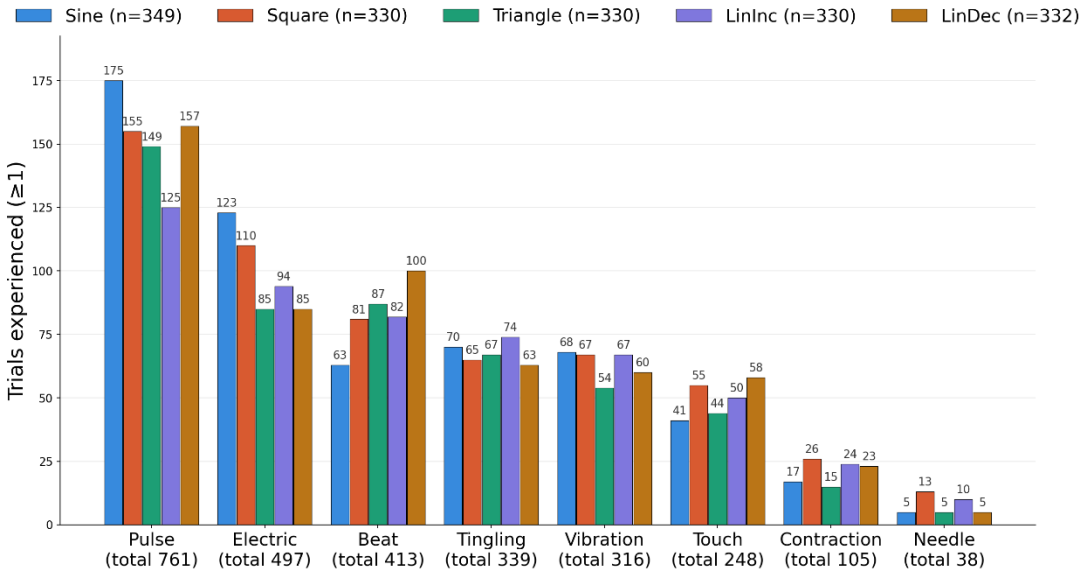


Figure 9: Recruitment rates across sensations highlight the influence of waveform shape on evoked sensation. The sine and square waveforms commonly evoke pulse and electric

sensations. The decrease waveform is able to evoke touch the most among the five waveforms; however, its rate of recruitment is still quite low (17.4%). Equally important is the rate of reduction waveform display, as triangular waveforms (besides linear decrease for pulses) evoke pulse and electric sensations at a much lower rate than non-triangular waveforms.

A secondary analysis can be performed to further separate the rate of evoked sensation by the overall amplitude the waveform was administered at. The majority of sensations were evoked at a higher rate across waveforms at the highest charge of stimulation. A clear outlier to this is the touch sensation, which was most commonly evoked at a non-high charge amplitude for every waveform besides linear increase. These findings suggest that there is a relationship between a waveforms associated charge and its rate of recruitment of certain sensations. Thus, an optimal charge can be determined for each waveform shape that presents the highest recruitment of desired

sensations, and a lower recruitment of undesired sensations. This optimal charge will vary between participants, due to the differences in threshold charge required to evoke a sensation.

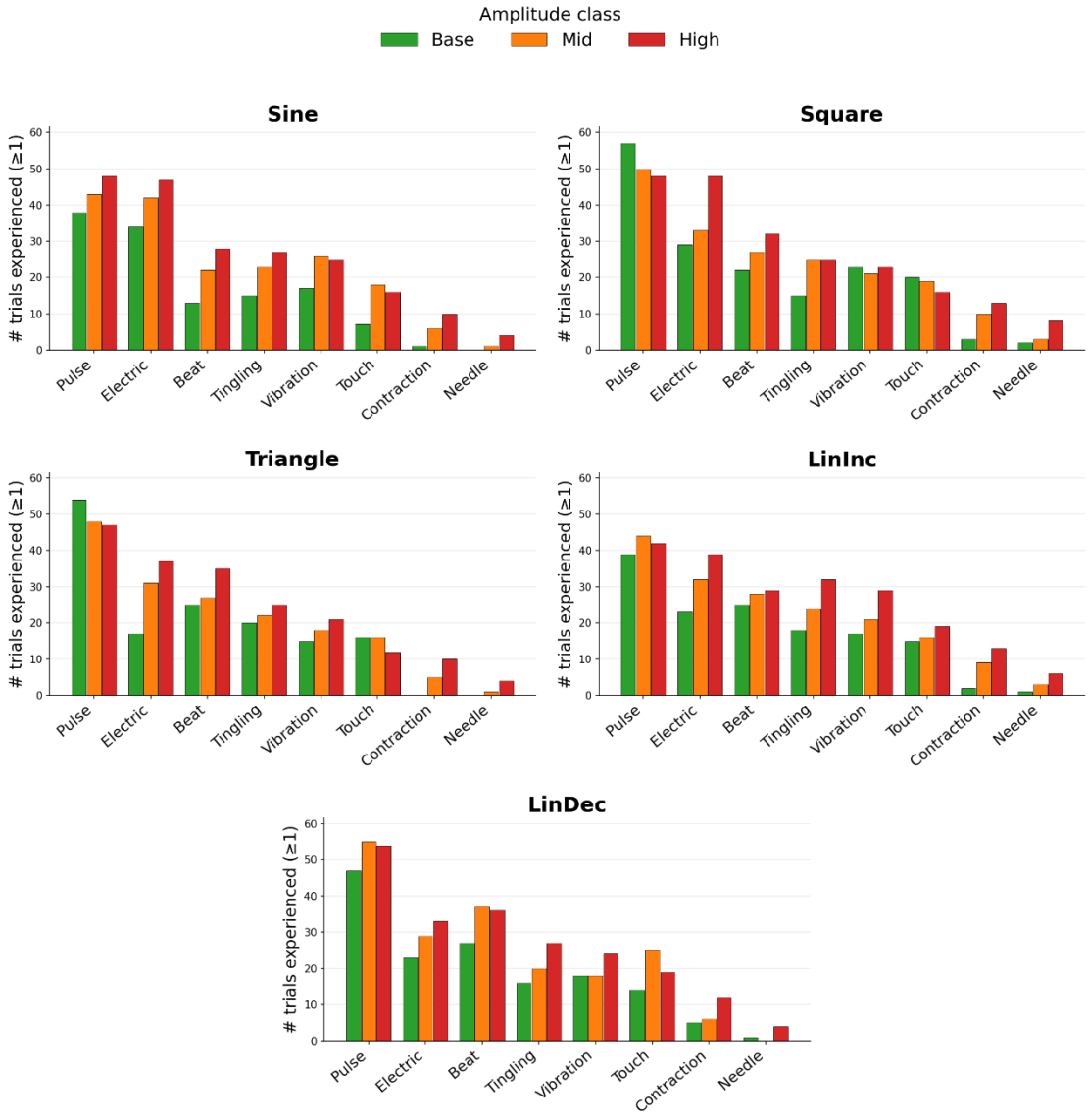


Figure 10: Recruitment rates split between waveform amplitude showcase distinct changes between charge levels. The touch sensation was recruited at the base or mid amplitudes for four out of the five waveforms, highlighting that charge-specific optimization can lead to a higher rate of recruitment for desired sensations. Similarly, the electric sensation was most commonly recruited at high amplitudes across all five waveforms and thus could be avoided through charge optimization.

Various unique areas of sensation are clearly presented when filtering average sensation location by sensation type, even when that averaging is done across all five different waveforms. Electric, pulse, tingling, and needle all showed average sensations that were able to escape the general palm area. The electric sensation evoked sensations across all fingers besides the pinky, and the needle sensation evoked sensations across the thumb consistently. The pulse sensation correlated with a distinct "line" of sensation from the middle of the palm to the tip of the middle finger, and the tingling sensation correlated with evoked sensations across a wide range from the tip of the thumb to the tip of the middle finger. Interestingly, touch, vibration, contraction, and beat sensations were only experienced on the palm of the hand. Further experimentation should investigate the driving factor behind why certain sensations are seemingly "region locked" at near threshold amplitudes to specific locations of the hand, even when the targeted nerve spans a much larger area. The relationship between average charge and average sensation area is again scattered. Pulse, with the highest charge, was experienced over the second lowest average area of the hand (3.4%), while tingling, tied with the second lowest charge, was experienced over the highest average area (4.6%). Various other mismatches between charge and area are present, with only contraction and vibration matching their expected position assuming a direct relationship. The combination of the charge associated with a sensation not correlating closely to its associated perceived intensity or area highlights the subjective nature of evoked sensation types, which is likely compounded in large studies. This also suggests that when a category, such as sensation

type, is averaged over various waveform shapes, the expected charge relationships become difficult to observe.

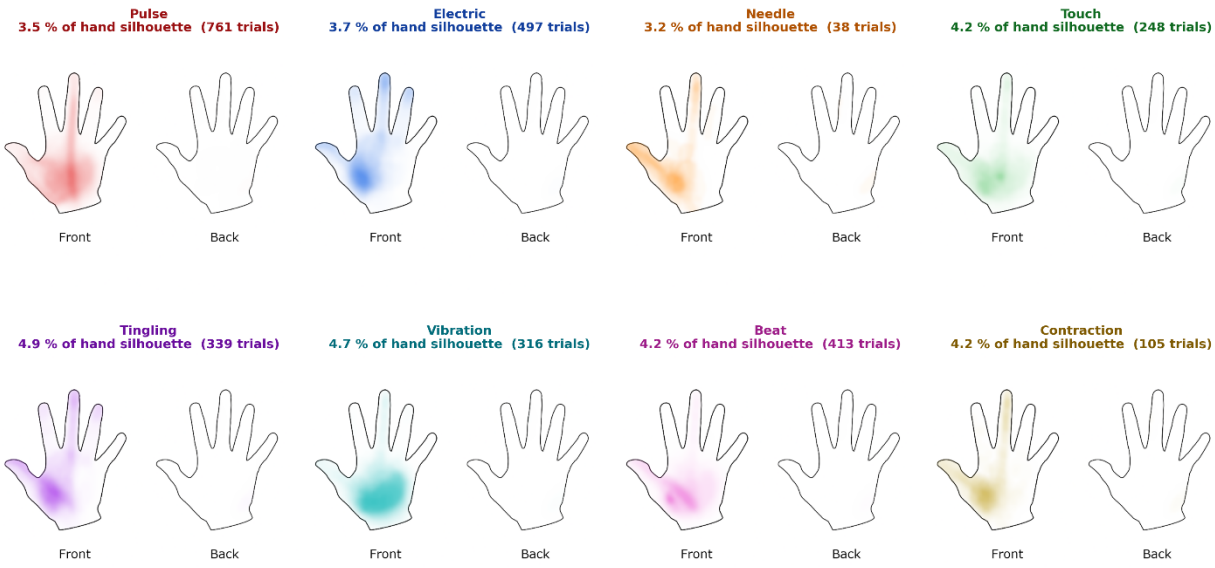


Figure 11: Grand average evoked area per sensation. The electric sensation was experienced across all fingers that correspond to the median nerve; the pulse waveform was experienced across a distinct sensation line from the palm across the middle finger; the needle sensation was consistently experienced on the thumb; while the tingling sensation was experienced across the highest average area. Touch, beat, vibration, and contraction sensations were only consistently experienced on the palm.

This subjective nature is apparent visually, as some sensations, such as pulse, visually display an overall shaded area that seems not to correlate with its associated average percentage of area shaded per trial. This discrepancy is a result of the wide range of subjective experiences that the specific sensation evoked, as for various participants that sensation correlated with a large area, and for others, the exact opposite was experienced. This is also a result of certain subjective adjectives correlating with different sensations across participants.

The effect of super-threshold stimulation on evoked sensations can also be drawn from this study. To fully characterize the super-threshold amplitudes of stimulation, the average charge value for each stimulation amplitude across waveforms is shown in Figure 12. Due to the difference in charge step size between waveform shapes, the overall charge difference between the waveforms increases as the overall amplitude increases. At the mid amplitude, the average square wave's associated charge becomes significantly different from the linear increase waveform ($p = 0.035$), and even more significantly different than the triangle waveform ($p = 0.010$). At the highest amplitude, the average square wave's associated charge significantly differs from the triangle ($p = 0.010$), linear increase ($p = 0.010$), and linear decrease waveforms ($p = 0.016$). The average sine wave's associated charge significantly differs from the triangle ($p = 0.016$) and linear increase ($p = 0.035$) waveforms at this amplitude as well. Thus, the rate of change for evoked sensation attributes of intensity and area are expected to increase at higher rates for the square and sine waveforms compared to the three triangular waveforms.

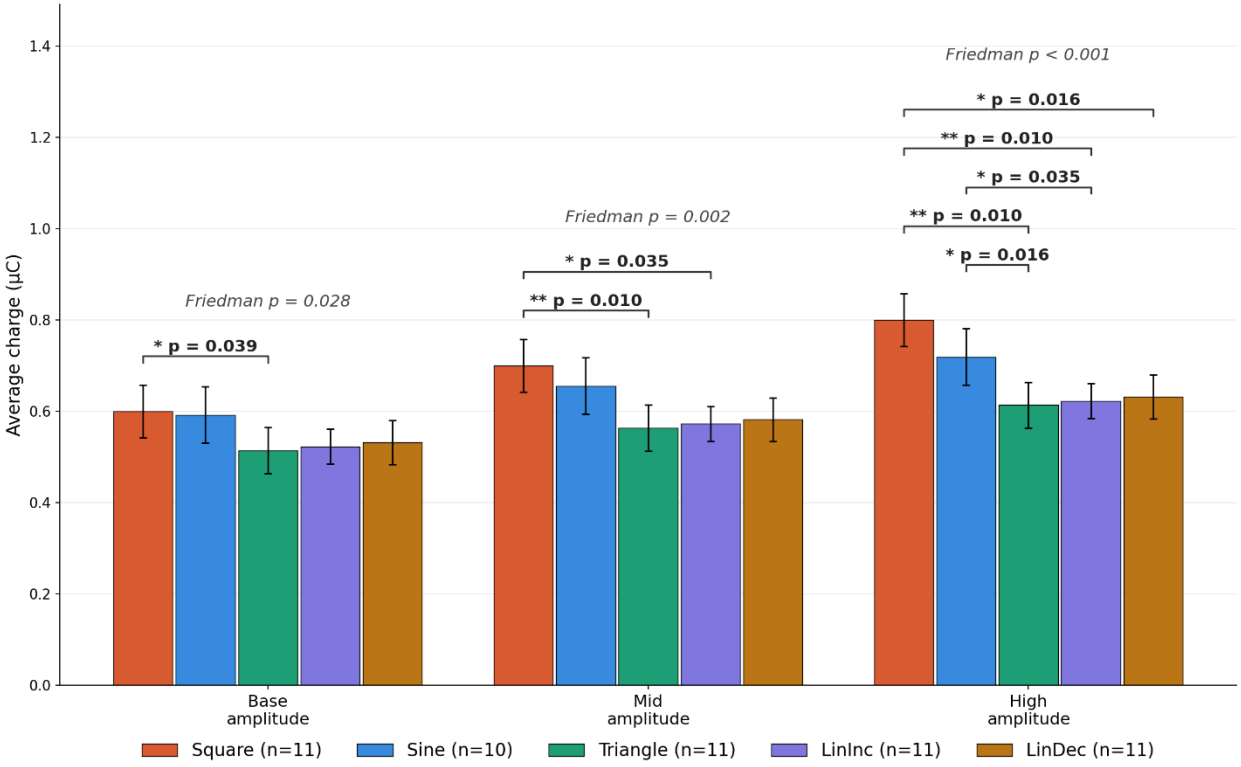


Figure 12: Charge imbalance between waveforms scales as amplitude increases. At the high amplitude, the square waveform is significantly higher than all of the triangular waveforms, and the sine wave is significantly higher than two of the three. With this in mind, we expect the square and sine waveforms to evoke intense sensations over a larger area of the hand than the triangular waveforms. Further, as the square waveform experiences the greatest change in charge from base to high amplitudes, it is expected to have the largest increase in both intensity and area.

As expected, a higher amplitude of stimulation, and therefore a higher overall charge, corresponded to a larger area of an evoked sensation on the hand across all five waveforms. The square waveform had the highest difference between base and high-amplitude areas, as expected, with the highest-amplitude waveform evoking sensation in an area 2.3x larger than the base amplitude. The sine waveform exhibits a smaller difference, with its highest amplitude waveform evoking a sensation area 1.74x larger than the base amplitude. Although this difference is smaller for the sine waveform, it should be noted that it evokes sensation in a larger area than the square

waveform for all three amplitude bases. Thus, the sine waveform is able to evoke a larger sensation area than the square waveform with a lower overall charge across three different amplitudes of stimulation. At the highest amplitude of stimulation, the sine waveform's associated charge is 0.081μC less than the square waveform while evoking sensation over a 0.5% larger area. These findings strengthen the conclusion that temporal attributes of a stimulus waveform can modulate evoked sensation area, as the sine waveform is able to evoke a notably larger area than the square waveform even with a (relative to this study) large separation in overall charge. However, as the charge between the two waveforms increased, the difference in area decreased from a 1.1% difference at the base amplitude to a 0.5% difference at the high amplitude, suggesting that the meaningful difference that a waveform's shape can present in regard to sensation area diminishes when the charge difference becomes extreme. As expected from the previous analyses, the three triangular waveforms evoke lower areas of sensation than the square and sine waveforms across all three amplitudes. However, they each still exhibit a clear increase in area as charge of stimulation increases, with the triangle, linear increase, and linear decrease evoked areas scaled by 1.72x, 1.909x, and 1.625x, respectively, from base to high amplitude. The hand heatmaps for each amplitude level across all five waveforms can be seen in Figure 13.

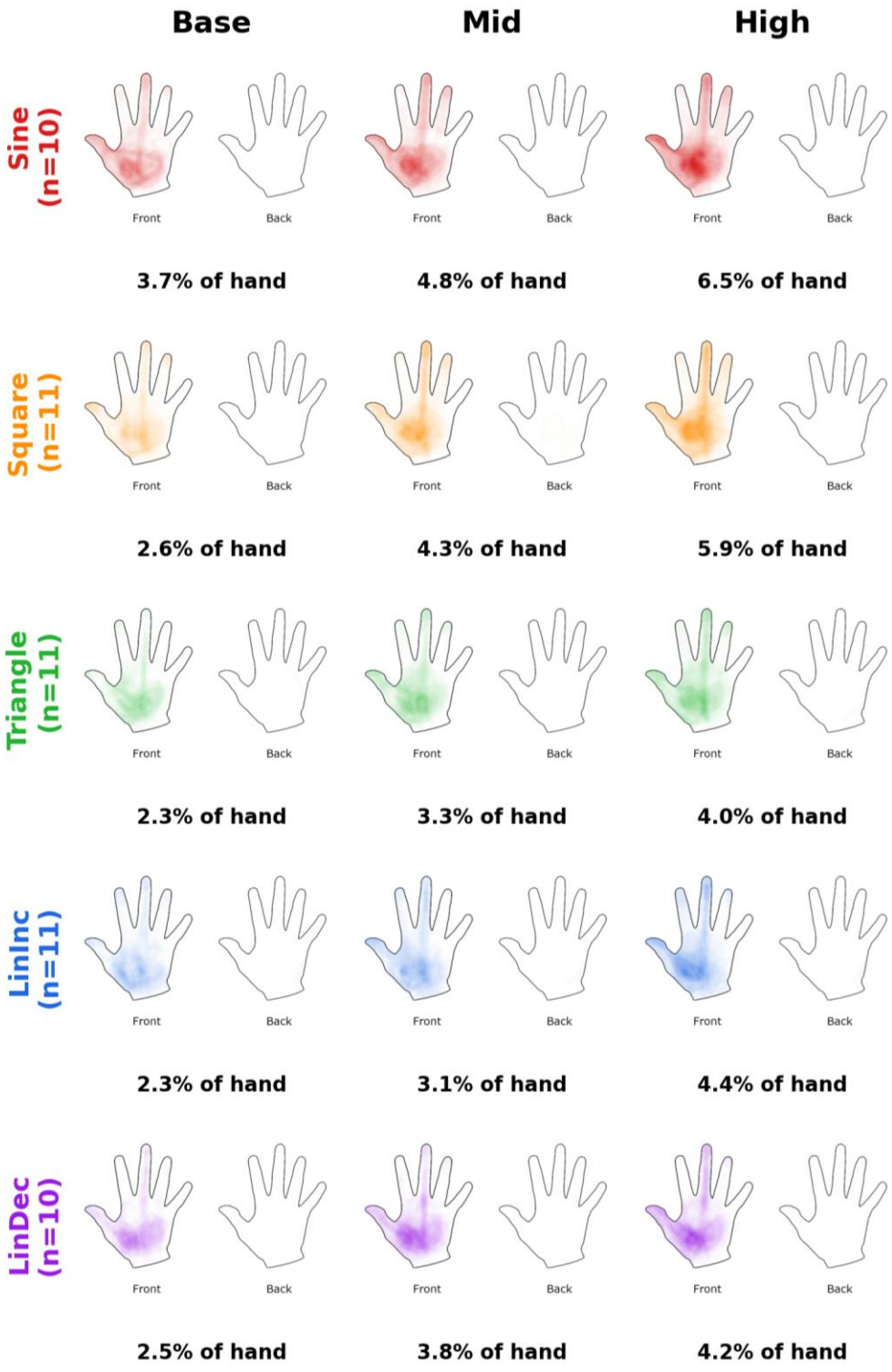


Figure 13: Grand average evoked sensation by waveform and amplitude. As amplitude increases, the overall area of the hand that experienced sensation increases regardless of stimulus waveform. This matches expected results, as the increase in overall charge causes a larger response.

Similarly, the increase in charge correlated to an increase in perceived sensation intensity as well. Interestingly, the sine waveform was able to evoke a more intense average sensation than the square waveform at their individual base amplitudes, even though the sine waveform had a (marginally) smaller charge. As the amplitudes of the waveforms increased, the square waveform returned to its expected position as the waveform that correlates to the highest intensity, likely due to the fact that the charge difference between the two waveforms increased. This suggests that waveform shape likely affects perceived intensity similarly to its effect on area, while being much more reactive to the charge differences between waveforms. The three triangular waveforms started and remained closely packed together at an intensity level much lower than the sine and square waveforms, although a clear increase in intensity is displayed as charge increases. The relationship between the amplitude of the stimulus waveform and the intensity of evoked sensation is shown in Figure 14.

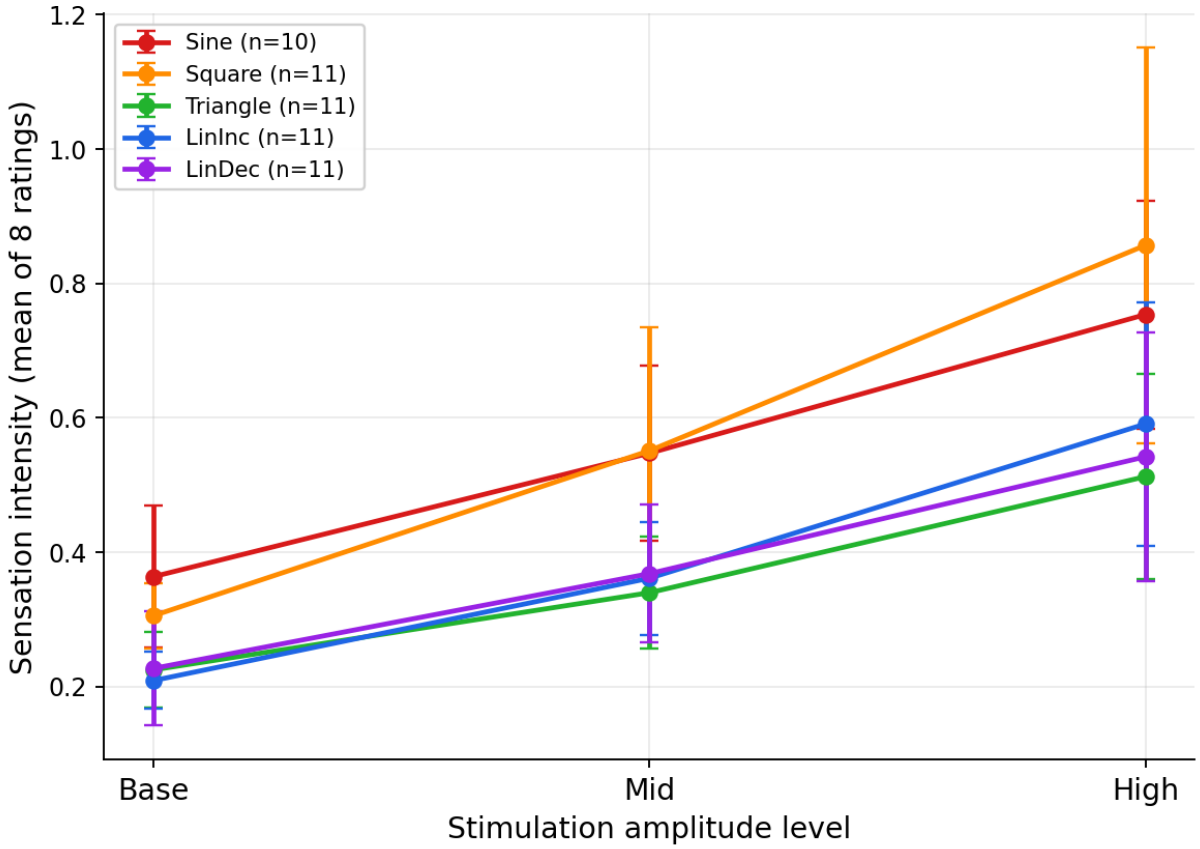


Figure 14: Average sensation intensity across amplitudes. Perceived intensity increases as amplitude increases across all waveforms, matching the expected response due to the associated increase in charge. Interestingly, the increase in amplitude does not affect each waveform the same, as the square waveform trends steeper than the sine.

Waveform Event Related Potential Comparisons

As previously discussed, a common method for analyzing EEG data is through the visual analysis of a stimulus's associated event-related potential (ERP). For research concerning evoked sensations, these ERPs are referred to as a somatosensory evoked potential (SEP). When generating these SEPs, it is important that the EEG data used is collected from relevant electrodes for the specific stimulus that is being analyzed, to reduce noise generated from off-target brain areas. Thus, careful consideration must be taken into the selection of which electrodes will be used.

Electrode Selection for SEPs

The stimuli applied are paramount to determining which electrodes are relevant for analyzing EEG data. This experiment utilizes electrical stimulation as its stimulus, which is delivered to the median nerve corresponding to the participant's left hand. The goal of the electrical stimulation is to evoke a sensation on the participant's hand, and thus the relevant electrodes are those that correspond to the sensorimotor cortex. These electrodes (based on a standard 10-20 montage) consist of the fronto-central (FC), central (C), centro-parietal (CP), and parietal (P) electrode rows. From this general electrode area, previous research has identified the C3/C4 electrodes to correspond directly to the right and left hand [31]. With these electrodes of interest in mind, SEPs were computed using various combinations of electrodes.

Electrode combinations were chosen based on both established literature concerning hand SEPs and through a search of which combinations yielded relevant significant events. Early extrema components (N20/P27) are highlighted by contralateral central-parietal electrodes, with CP3/CP4 specifically containing important EEG information concerning upper limb evoked sensations [32]. Mid extrema components (N140) have been shown to be captured reasonably accurately through Cz/FCz electrodes and contralateral C/CP electrodes [33]. Late extrema components (P200/P300) can be accurately captured through the Centro-parietal midline (CPz/Pz/Cz) [34]. With these relationships between electrodes and SEP components, conclusions can be drawn between various stimuli and SEP components with confidence that the EEG data observed is directly relevant to the stimuli applied.

These significant electrodes can be divided into specific regions of interest (ROI) for which SEPs can be averaged over. ROIs were determined based on the previously discussed correlations

to the somatosensory cortex and which ROIs produced significant SEPs. Within this analysis, significance is attributed to events where all five group means between the waveforms differ. The significant SEPs considered consisted of N70, P100, N140, P200, and P300. The final ROIs chosen were; Contralateral Centro-parietal (CP4, CP2, CP6, C4, P4), and Contralateral parietal (P4, P6, P8, PO4). In addition to these two ROIs that were found to be significant for this experiment, three additional ROIs will be included for analysis as well. These additional ROIs will match the three ROIs included by Alibou et al. in their case study investigating TENs and EEG data [34]. These three ROIs are central frontal, contralateral parietal, and superior parietal. This study administered TENS to the left hand, whereas Alibou et al. administered TENS to the right hand. Thus, the specific contralateral electrodes will be flipped for consistency, ensuring that the electrodes used are still contralateral to the hand whose sensations are evoked. The specific electrodes corresponding to each group can be visualized in Figure 15. The addition of these ROIs will provide a baseline to compare our SEPs to, ideally drawing similar conclusions between relevant latency events of P50, P150, and P300 [35].

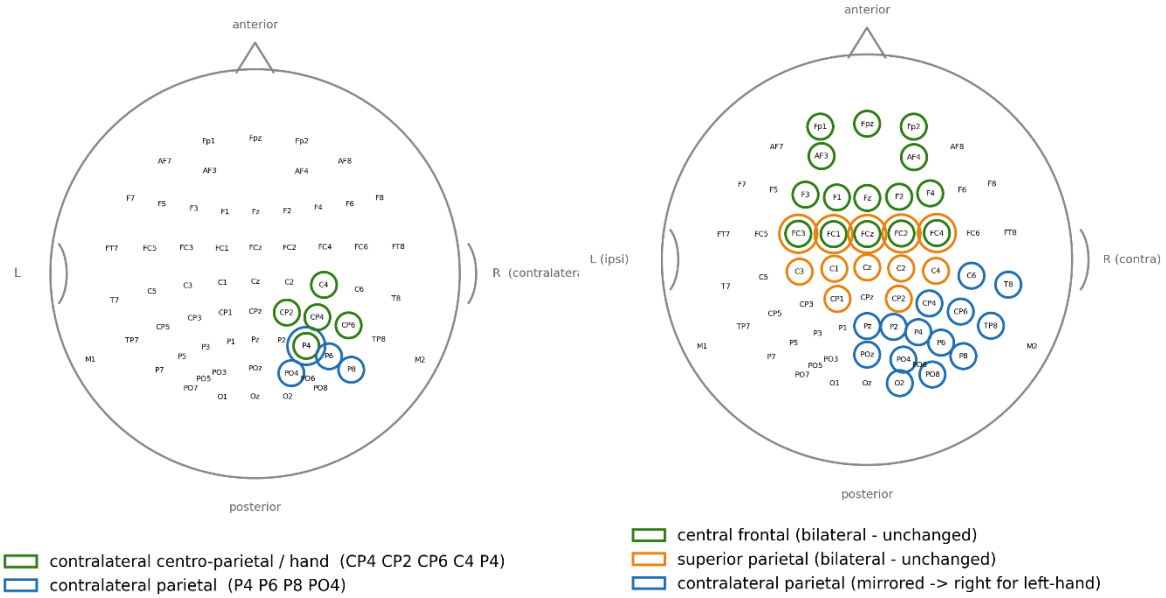


Figure 15: (Left) Defined ROIs based on electrode sweep. ROIs include Centro-parietal (CP4, CP2, CP6, CP4, P4) and parietal (P4, P6, P8, PO4). (Right) Literature ROIs used for comparison across studies. Note the contralateral-parietal ROI is mirrored across the midline of the headcap to account for the differences in stimulation side between studies (right vs left). The ROIs selected are central frontal (Fp1, Fpz, Fp2, AF3, AF4, F3, F1, Fz, F2, F4, Fc3, Fcz, Fc2, Fc4), superior parietal (Fc3, Fcz, Fc2, Fc4, C3, C1, Cz, C2, C4, Cp1, Cp2), and parietal (C6, T8, Cp4, Cp6, Tp8, Pz, P2, P4, P6, P8, Poz, Po4, Po8, O2).

SEP Comparison - Between Stimuli Class

SEPs were first analyzed using data that combined epochs by the waveform class, rather than by the amplitudes within the waveform class. This provided a baseline for comparison across waveforms with a large dataset consisting of 150 epochs per waveform per participant. Each ROI was inspected to discover relevant SEPs across all waveforms, as well as areas of significant differences between the waveform means that may provide insight to correlations between the

different subjective experiences of each waveform through their associated differences in neural response. These significant differences can come in two manners; significant difference in mean amplitude, and significant difference in latency between expected events. Significant differences in mean amplitudes were calculated using windowed means around the SEP of interest, to provide ample temporal information to reduce false significance.

Visual analysis of the centroparietal ROI revealed occurrences of various SEP events (Figure 16), some of which slightly differed in latency from their expected positions. Specifically, the P300 SEP appears to peak in the upper P3 range across all classes of waveforms. Although visually insignificant, the mean amplitude of the N140 SEP significantly differed between the five waveforms ($p = 0.005$), whereas the amplitude of the P300 SEP significantly differed ($p = 0.004$) as well. The significant difference in amplitude between waveforms for the N140 SEP suggests that the associated area a stimulus evokes can modulate neural activity significantly, as this event is expected to be highly sensitive to the physical location of the evoked sensation [32]. The significant difference in amplitude between waveforms for the P300 SEP is tied to neural activity that correlated with the participants' evaluation of how a certain stimulus felt [33], reinforcing the subjective reporting that the temporal differences in the stimuli presented notably different intensities, sensation areas, and sensations to the participant. The amplitude hierarchy between waveforms in both the N140 and P300 amplitudes almost perfectly maps the waveform that evoked the largest area of sensation. The waveform found to have the highest amplitude for both SEPs, sine, was found to evoke the largest area of evoked sensation on average by participants. Sine is closely followed in amplitude within these SEPs by the square waveform, which was found to evoke the second largest area of sensation, and the most intense sensation on average. Further, these SEPs suggest that charge is not the core influencing factor behind neural amplitude, as the

square waveform was found to have a lower neural amplitude at these SEPs than the sine waveform, even though the square waveform had an average charge 0.044 μC higher than the sine waveform. The three triangular waveforms, which all averaged lower charge and evoked significantly less intense sensations within a smaller area of the hand, present much lower amplitudes. These findings suggest that the driver behind differences in neural activity during evoked artificial sensation is not purely the charge contained by a stimulus, but a combination of the charge and waveform shape. Although weaker, this data could suggest that the area over which a sensation is experienced is the main outcome of high neural activity, as the sine waveform (1st in area, 2nd in intensity) is found to correlate with a higher neural amplitude than the square waveform (2nd in area, 1st in intensity). This suggestion would align with the waveform shape being the main driver of higher neural amplitudes, rather than charge (assuming the charge difference between waveforms is relatively small).

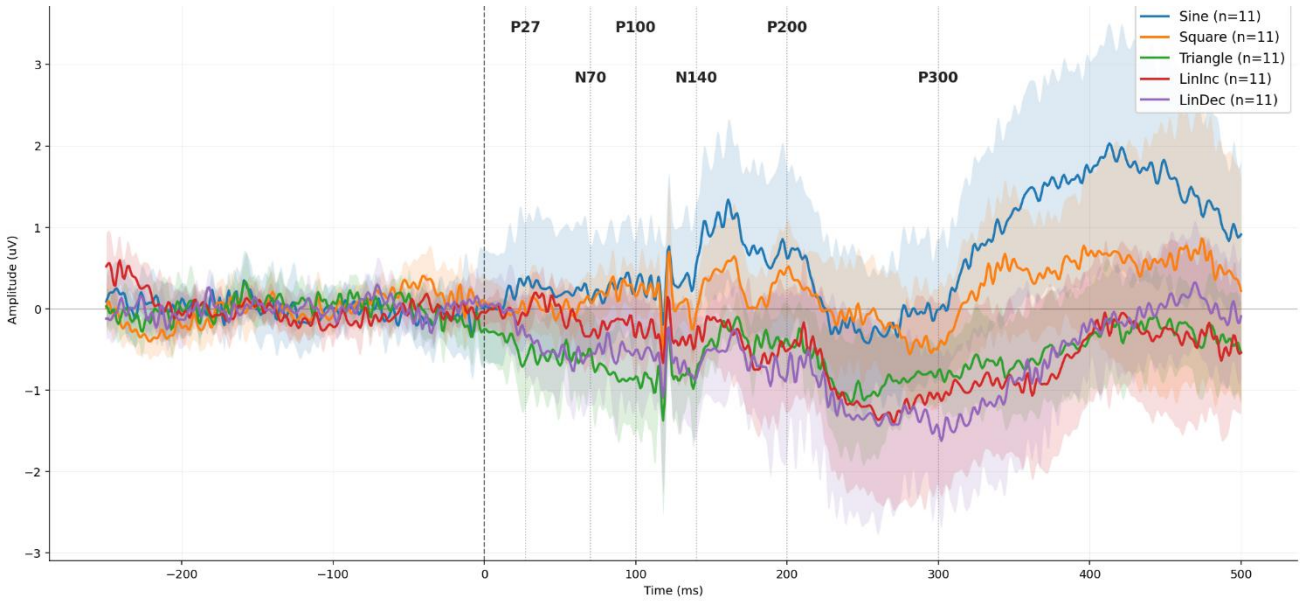


Figure 16: Centro-parietal ROI SEPs across all waveforms. Shaded locations indicate a 90% confidence interval. Significant SEPs occur between waveform means at N140 ($p = 0.005$) and P300 ($p = 0.004$). These differences in means suggest that the area a sensation occurs (N140) and

the evaluation of the sensation (P300) both have significant effects on temporal neural activity. Throughout the entirety of the post stimulus SEP, the sine waveform exhibited the highest neural amplitude, closely followed by square. This likely is caused by the high combination of intensity and area that both of these waveforms were able to evoke compared to the three triangular waveforms.

Analysis of the parietal ROI reveals similar SEPs as the centroparietal ROI (Figure 17). The P200 amplitude is significant between the five classes ($p = 0.035$), however the ordering at this point makes a notable shift. Slightly after this event, the square waveform showcases a higher amplitude than sine, being able to recruit more neural activation even with a lower overall area. This ROI is able to highlight the short amplitude swap found in the centroparietal ROI at the same timing, extending the amplitude mismatch to a significant difference. Further from the onset of stimulation, the P300 amplitude significantly differed between the five classes ($p = 0.01$). Around this timing, the sine waveform regains its place as the waveform with the highest amplitude, as the amplitude created by the square waveform decreased while the amplitude associated with the sine waveform increased. These findings suggest that the difference in neural recruitment between the shape and intensity of an electrical stimulus may correlate with temporal amplitude changes within an SEP, which can further be tied to the expected area and intensity of an evoked sensation.

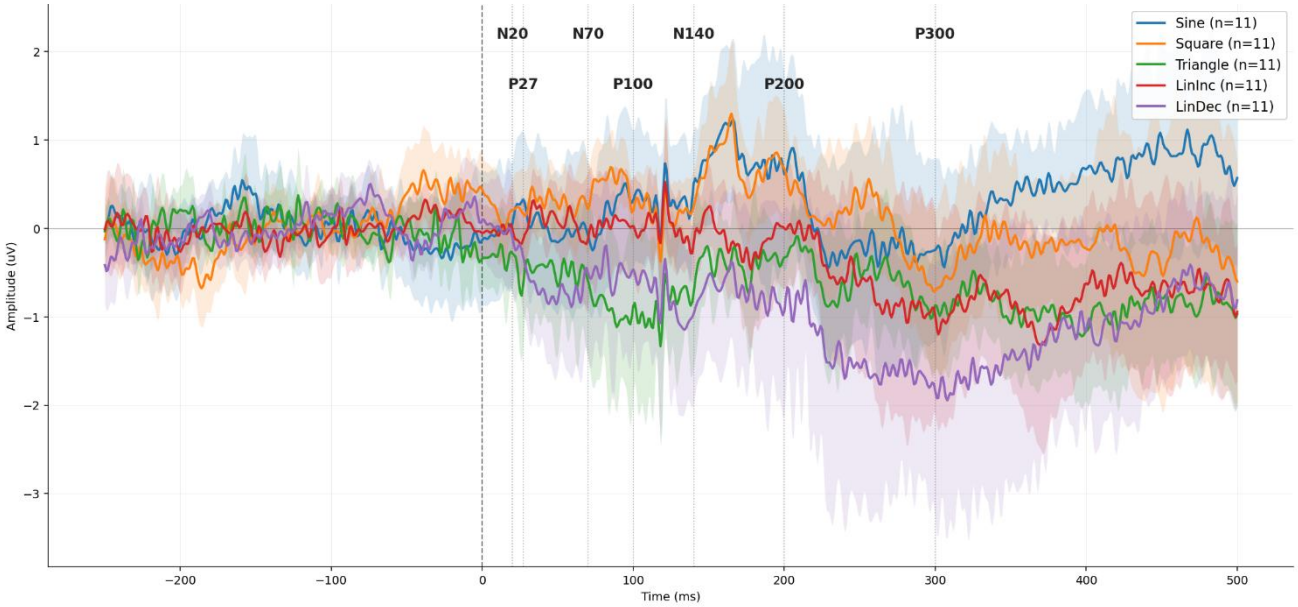


Figure 17: Parietal ROI SEPs Across all Waveforms. Shaded locations indicate a 90% Confidence Interval. Significant SEPs occur across waveform means at P200 ($p = 0.035$) and P300 ($p = 0.01$). The square waveform is found to have the highest amplitude around the P200 event, until the sine waveform reaches its expected position around the P300 event. The three triangular waveforms exhibit consistently smaller neural amplitudes in comparison to non-triangular waveforms.

Similarities between the central frontal, superior parietal, and contralateral parietal SEPs found in this study and those in Alibou et al. are surprisingly limited (Figure 18). Alibou et al. notes meaningful potentials during TENS of the median nerve, including P60 within the contralateral parietal ROI, N60 and P150 within the central frontal ROI, as well as P300 within the superior parietal ROI [34]. These meaningful potentials are not present in the central frontal and contralateral parietal ROIs within this study. However, the superior parietal ROI bears comparison, as it similarly showcases a P300 peak (although much later) across all five classes of waveform. The general lack of similar SEPs between studies suggests that these SEPs may be subject-specific and thus will greatly diminish when averaged across larger datasets, as the findings in Alibou et al. were based on SEPs generated by a single subject. The central frontal ROI

interestingly showcases that in certain brain regions, a lower charge can result in a higher neural amplitude, as the three triangular waveforms are each higher than the square and sine waveforms.

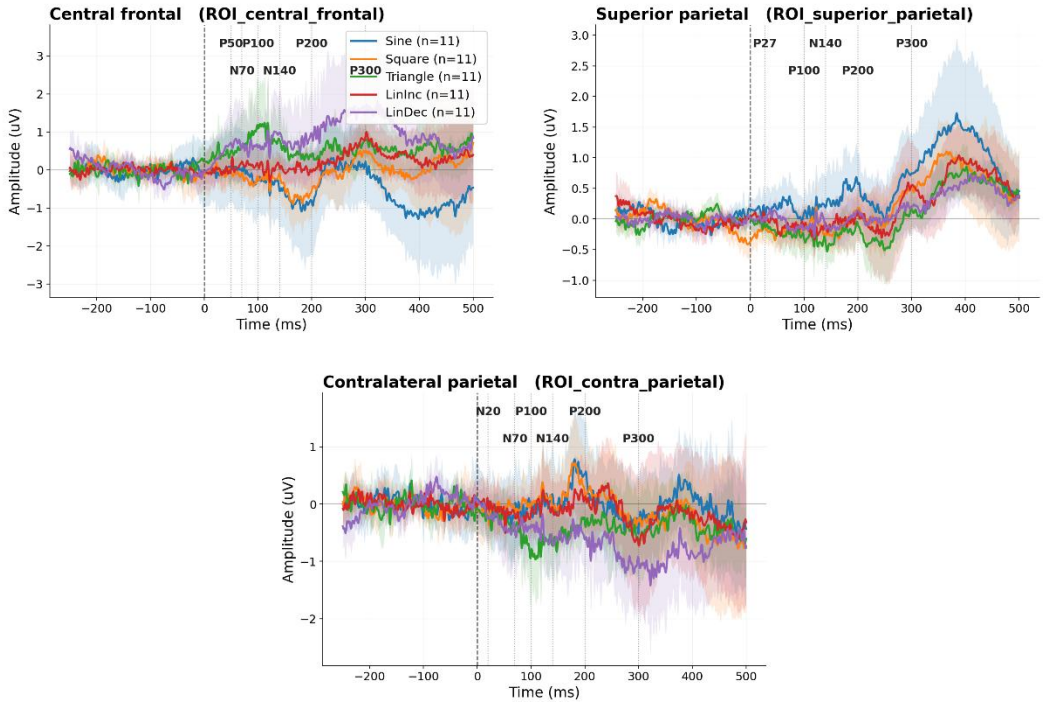


Figure 18: Literature ROI SEPs. Comparisons between SEPs within this study and available literature are limited, with no comparable events in either of the central frontal or contralateral parietal ROIs. The only similar SEP was found in the superior parietal ROI at the P300 event, although within this study the SEP occurred much later in the P300 window than expected.

SEP Comparison - Between Stimuli Classes Conclusions

The analysis of SEPs within the central parietal and parietal ROIs reveals various relationships between stimulus waveform and EEG data. Generally, the waveform with the highest amplitude of neural activity was the waveform that evoked the largest area of sensation on the hand. The sine and square waves, which were able to evoke the first and second largest area of sensation and intensity of sensation, were observed to evoke a significantly higher average amplitude of neural activity than the three triangular waveforms post-stimulation ($p = 0.002$,

Centro-parietal ROI). The triangular waveforms were observed to have similarly small amplitudes due to the fact that they evoked subtle sensations over a small region of the hand in comparison to non-triangular waveforms. These findings are driven by both the charge and shape of the stimulus waveforms. The typical ordering between neural amplitudes closely followed the hierarchy of average charge presented by each waveform. Notably, the square waveform was found to result in a lower neural amplitude than the sine waveform, with the only exception being around 250 ms post-stimulation, even with a 0.044 μC higher average charge. These findings support the importance of temporal information carried by a stimulus waveform and showcase the consistent effect that this information can have on neural response across a diverse participant base. Further, the sine waveform maintained a higher amplitude than the square waveform in the P300 region across both ROIs (centroparietal = +0.73uV, parietal = +0.63uV), suggesting that a larger area of evoked sensation, likely driven by the waveform's shape, correlates to increased neural activation in comparison to a higher intensity of sensation during the evaluation stage of sensory processing. The square waveform corresponds to the highest neural amplitude for a short period (around 250ms), suggesting that the relationship charge and shape have with neural amplitudes has a consistent temporal component. This temporal component could be used for extracting expected sensation intensities and areas in situations where participants are not able to provide subjective feedback. The overall result of sine consistently correlating to a higher neural amplitude could be the combination of both its shape and charge, and thus a combination of the intensity and the area its evoked sensations were experienced, rather than an assertion that one is more important than the other. To determine if the sine waveform has a larger combined intensity and area than the square waveform, a combined score that equally weights the importance of each category must be

created. This combined score will represent the total expected "drive" a waveform is associated with. To calculate this drive, the following formula was used

$$D_w = 0.5\, z(Area_w) + 0.5\, z(Intensity_w) \quad \text{(Eq. 1)}$$
$$z(x_w) = \frac{x_w - \mu_x}{\sigma}$$

The norm was calculated for each category using z-score normalization across the five waveforms. This equation results in the sine waveform having a drive of 1.27, and the square waveform having a drive of 1.09 on a relative scale between waveforms (linear decrease, linear increase, and triangle have drive factors of -0.71, -0.78, and -0.88, respectively). Thus, the neural amplitude observed is likely driven by this combination of sensation intensity and area, as the sine waveform has a higher overall drive based on the two categories in comparison with the square waveform. The temporal changes observed may still have relationships to certain features of the stimulus and its associated evoked sensation, but concrete information regarding whether one or the other category produces noticeably different activation cannot be drawn from a simple amplitude analysis. A deeper analysis of these two waveforms should be completed to produce a robust conclusion on this relationship.

SEP Comparison - Between Stimuli Intensities

SEPs were further separated based on amplitude within a certain waveform class, investigating possible neural differences. Within this section, SEPs from the parietal ROI will be utilized to showcase differences between the three charge amplitudes, due to their ability to highlight their associated significant differences. Within the square waveform class, significant

asynchrony in SEP timings occurs at the N140 latency within the parietal ROI (Figure 19). The largest visual difference occurs between the highest amplitude and the base amplitude, as the base amplitude's P100 peak occurs around 40 ms before the high amplitude's. The mid amplitude interestingly does not exhibit a positive peak post-stimulus until 150 ms. From 100ms to 200 ms post-stimulation, the highest amplitude creates a notably higher amplitude of neural response from the base and mid amplitudes. This high neural amplitude during the highest stimulation amplitude of the square wave could explain why the square waveform evoked the most intense subjective experience for certain sensations. Interestingly, within this same region, the base amplitude exhibits a higher amplitude of neural activity than the mid amplitude. This is not expected, as the mid amplitude is able to evoke more intense sensations over a larger area of the hand than the base amplitude. Within the 300ms to 400ms period, the expected order of high, mid, and base amplitudes occurs with regard to the neural amplitudes. This matches the sensory processing stage at this time, as during this period the neural activity will be most closely related to the participant's subjective experience of the stimulus, rather than a purely biophysical response to the stimulus itself.

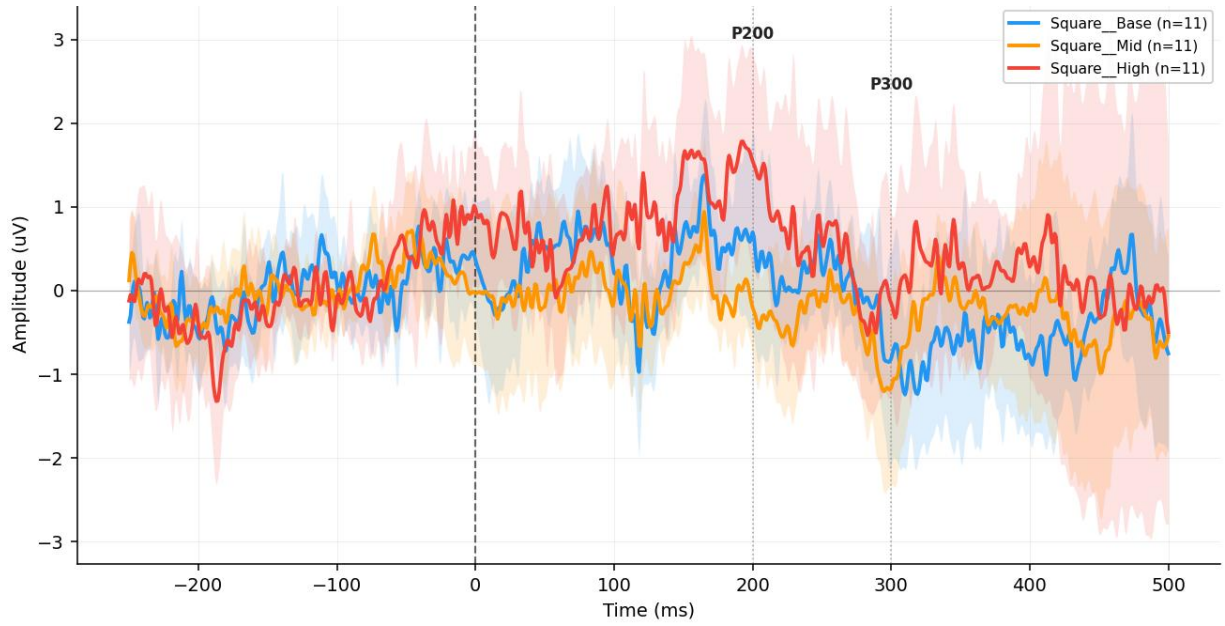


Figure 19: Parietal ROI SEPs for each amplitude of the square waveform. The three amplitudes present distinctly different neural amplitudes post-stimulus. Interestingly, the base amplitude is seen to evoke a higher amplitude than the mid amplitude, despite evoking a less intense sensation over a smaller area. At and past the P300 event, the expected ordering of high, mid, and base is observed.

The sine waveform displays similar ordering between charge amplitudes as the square waveform, with its order typically matching the expected amplitudes from the stimulation onset to 400ms post within the parietal ROI (Figure 20). Interestingly, the difference in amplitude between the base and high amplitudes for the square waveform is smaller for every component window (P50, N70, P100, N140, P200, P300). This difference is contrary to the subjective reporting, as the high-amplitude square waveform was reported to evoke a sensation 2.77x more intense over a 2.3x larger area than its base amplitude, whereas the high-amplitude sine waveform was reported to evoke a sensation 2.08x more intense over a 1.74x larger area than its base amplitude. The larger increase in intensity and area between the base and high amplitudes of the square waveform did not correlate with a larger difference in neural activity between the two classes when compared to the sine waveform. This could result from the sine wave evoking sensation over a larger area of

the hand at each stimulation amplitude, resulting in the neural response being on a larger scale than the square waveforms.

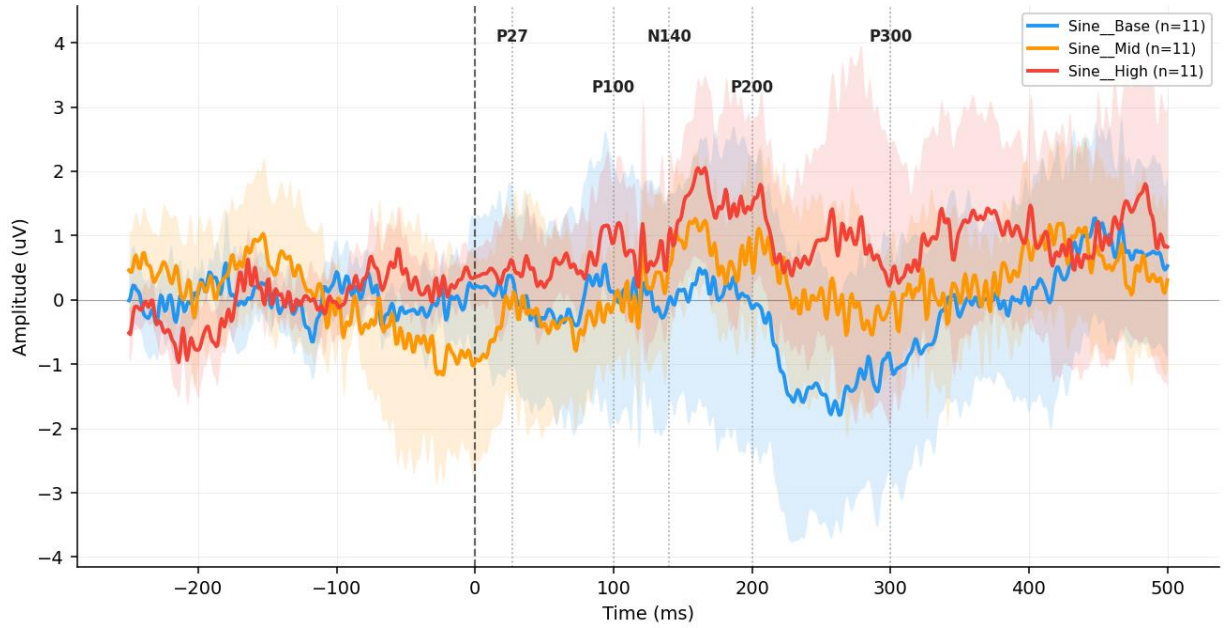


Figure 20: Parietal ROI SEPs for each amplitude of the sine waveform. Similar to the square waveform, the three amplitudes display a clear difference in associated neural amplitudes. After around 100 ms post-stimulation, the expected order between amplitudes occurs and is maintained for the rest of the SEP. Interestingly, the neural amplitude difference between the high and base amplitudes for the sine wave is higher than for the square wave, despite the difference between their evoked sensations being smaller.

The three triangular waveforms showcase a notably different trend in their neural responses to the three different charges of stimulation compared to the square and sine waveforms (Figure 21). Between the three amplitudes across all three triangular waveforms, no visual difference in neural amplitude was present in or around the P100 and P300 events. Interestingly, the only notable visual difference present is for the linear decrease waveform, as the base amplitude is notably smaller than the others around 150ms post-stimulation. This highlights the limited information visual analysis of ERPs can provide for evoked sensations that offer limited perception between the intensity and the area of an evoked sensation. All three waveform amplitudes were notably

lower in overall neural amplitude at similar post-stimulus events than those of the non-triangular waveforms. This likely further explains why the triangular waveforms resulted in a lower overall intensity and area of sensation during self-reporting, as their low charge at all three stimulation amplitudes (due to the smallest step size) did not differ enough for significant neural differences. However, the sine waveforms' base and mid amplitudes showcase notable differences even though they have a smaller charge difference than the triangular waveforms between base and high, highlighting the influence of shape as well.

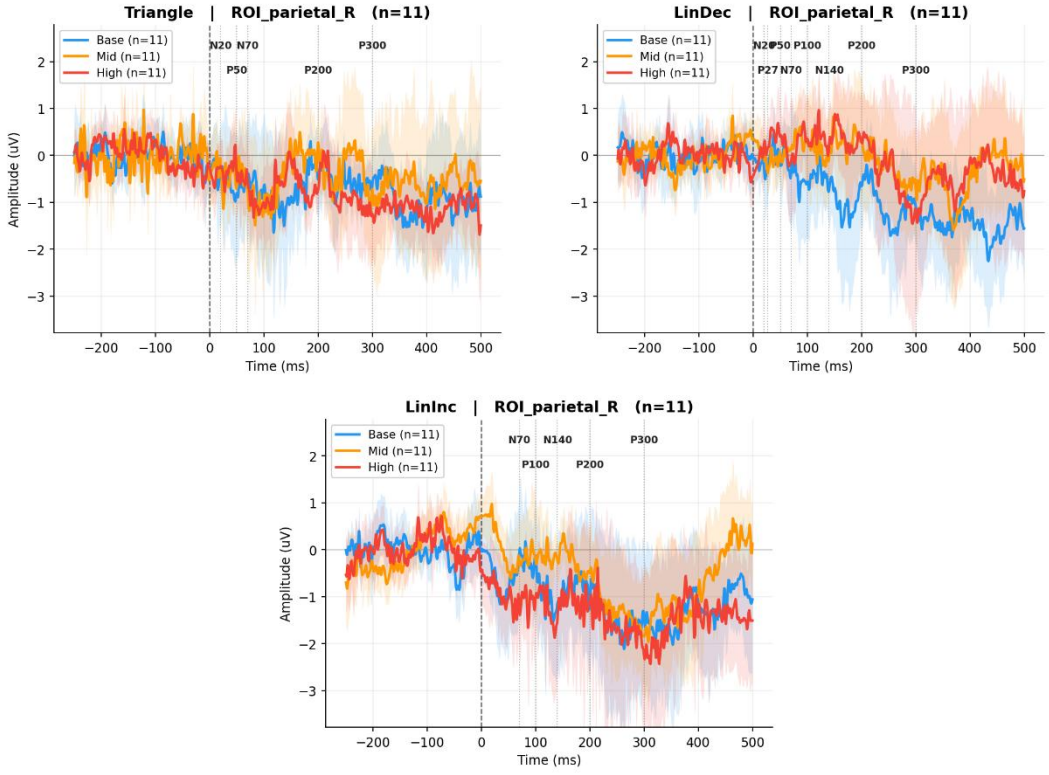


Figure 21: Parietal ROI SEPs for each amplitude of the three triangular waveforms. The different amplitudes provide no consistent visual difference between the three different waveforms, likely due to the small overall intensity and area of sensation evoked regardless of stimulation amplitude. Across the three waveforms, the only visual difference is in the base amplitude compared to the others regarding the linear decrease waveform.

Frequency Analysis of ROIs

ERSPs showcased significant synchronization in the theta and alpha bands across all five waveforms, as expected from previous literature [34]. This significant synchronization stretches from the stimulation event to the end of the epoch across nearly all of the ERSPs. Event-related synchronization (ERS) increased in magnitude and spectral range over time within the Superior Parietal ROI across all five waveforms. The triangular waveform, interestingly, resulted in ERS in the alpha band 100 ms post-stimulation in both the central frontal and contralateral parietal ROIs. Event-related desynchronizations (ERDs) varied greatly based on which waveform was under analysis, although it was observed to occur for each waveform. ERD occurred for the sine waveform within the beta band of the superior parietal ROI consistently across the post-stimulus period. A similar ERD occurred for the square waveform within the high beta band for the contralateral parietal ROI. The triangular waveform showed a robust ERD within the beta band across all three ROIs, while the linear increase and linear decrease waveforms exhibited weaker, less consistent beta ERD in the central frontal ROI. The ERSPs are thresholded against their own baseline, and thus comparisons between waveforms are qualitative and were not tested statistically against each other. Effects within 85 ms of the stimulation event are not considered, as the 150 ms analysis window overlaps with the 10 ms stimulation artifact blank.

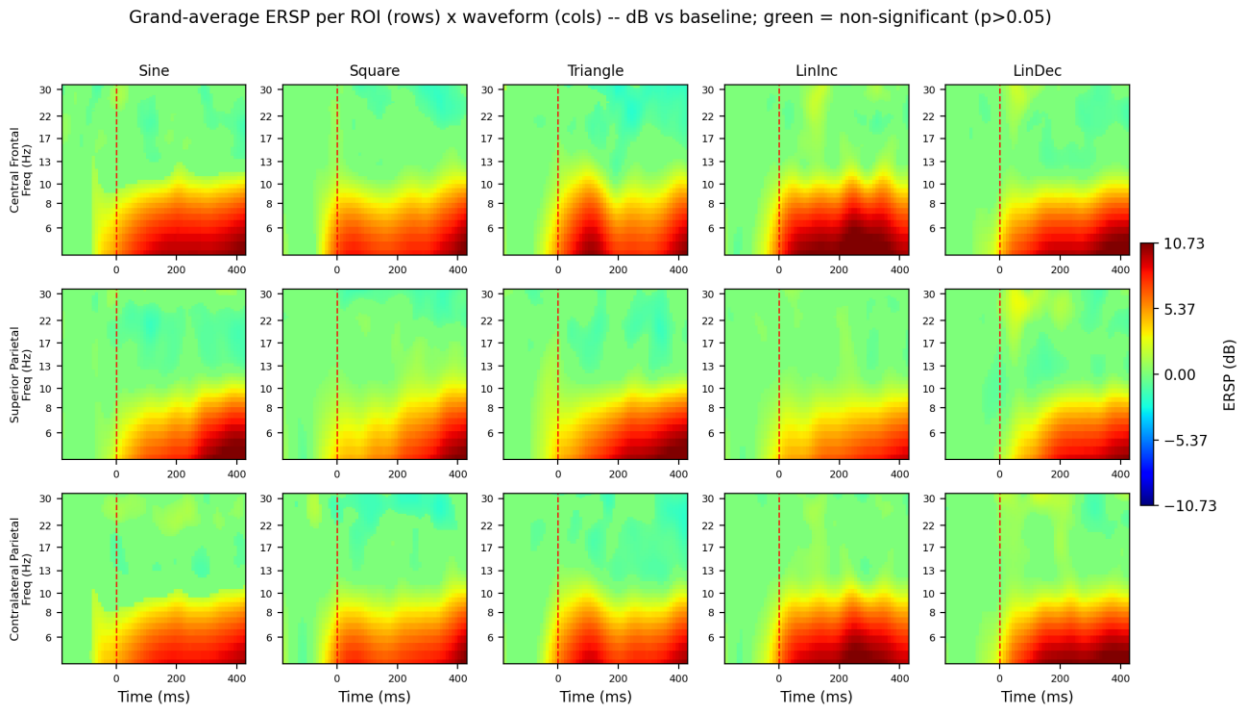


Figure 22: ERSPs computed over literature ROIs. Each ERSP exhibited ERS across the theta and alpha bands for each waveform, consistent with literature. ERDs were scattered across the ROI and waveform combinations, but occurred within the beta band for each waveform in at least one of the selected ROIs.

Two-Alternative Forced Choice

The two-alternative forced-choice task reveals clear consistency among participants in discriminating between waveforms, even when they are blinded to the waveform being sent. On average, participants consistently identified a preferred waveform within a set of two with an accuracy of 83.5% (± 15.82%), significantly higher than the chance level of 68.29% ($p < 0.001$). The theoretical chance level of 68.29% exceeds the intuitive 50% threshold due to the "race-to-five" task structure used. Within this task, the waveform set ends when one of the two waveforms is selected as preferred five times and thus receives 5 out of the N total selections (with $N \in 5$, ...

, 9). Under the null hypothesis of random selection ($p = 0.5$ per trial), set lengths are distributed according to a negative binomial with parameters $k = 5$, $p = 0.5$, yielding an expected within-set accuracy of 68.29%. However, this accuracy did not correlate with participants' agreement on which waveform was the most preferred. Linear decrease was found to be the most preferred waveform out of the five, but was only selected by 40% of the participants. This mismatch between participants' accuracy in distinguishing waveforms and their disagreement over their preferred waveform highlights the importance of using subject-specific stimulation parameters for electrical neuromodulation. Interestingly, linear decrease was not reported to have the highest mean “touch” sensation value among the five waveforms in the previous task; however, it was still chosen as the most preferred waveform. The conclusion that the linear decrease waveform was the most preferred waveform matches the subjective reporting by Collu et al., as in his study, linear decrease was found to have the highest average touch value during subjective reporting. This mismatch between this study's subjective reporting and the reporting in Collu et al., while reaching the same conclusion through an alternative forced-choice task, showcases the inherent inconsistency created through subjective reporting.

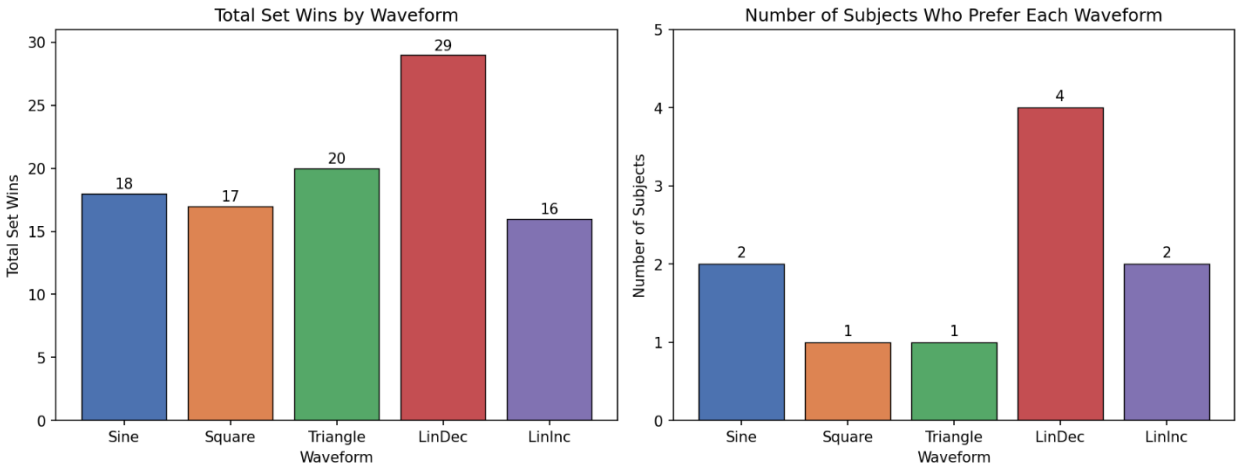


Figure 23: (Left) Total Set Wins by Waveform. The two highest win counts are both from triangular waveforms, highlighting the possibility that participants typically preferred sensations that were perceived as less intense over a smaller area. (Right) Total Number of Subjects who Prefer Each Waveform. The waveform with the highest set win was also chosen as the most preferred waveform. Outside of this consistency, the correlation between waveform set wins and subject preference is scattered. This highlights the inter-subjective variation between subjective experiences of the same stimuli.

The figure clearly illustrates the aforementioned mismatch: the linear increase waveform, which was chosen as preferred in only 40% of its sets, was the most preferred waveform for 20% of the participants. This ratio of set preference to most-preferred is much higher than for the triangle waveform, where it was chosen as preferred in 50% of its sets but as the most-preferred waveform for only 10% of participants. The combination of the participant's intra-set preferences with the varied overall most-preferred waveform highlights participants' ability to discriminate between waveforms while having significantly different subjective experiences. Overall, 65% of preferred sets contained triangular waveforms, and 70% of participants concluded the task with a triangular waveform being selected as most preferred. These results suggest that a combination of the triangular waveform's shape and low average charge results in a sensation that is found to be preferred to square or sine waveforms for the majority of participants.

This intriguing subjective consistency in intra-set preference raised an objective question: could the electrophysiological signatures that influence the participant's decision be detected in EEG data? In a sense, these signatures would be an objective representation of perceived naturalness, as the participant chose their preferred waveform as the one that felt most natural within the set. Visually, ERPs across the contralateral centroparietal and contralateral parietal ROIs showcase clear separation between the unchosen and chosen classes around the P200 and P300 events, with unchosen clearly exhibiting a higher amplitude. This suggests the waveform that was selected as more natural typically correlated with a lower amplitude of neural activity, and thus a lower perceived intensity and area of stimulation [33]. The lower amplitude of neural activity would typically correlate to a triangular waveform as opposed to a square or sine waveform as revealed through the previous SEP analysis. This matches the alternative forced choice task results, as the triangular waveforms won 65% of sets and were selected as most preferred by 70% of participants.

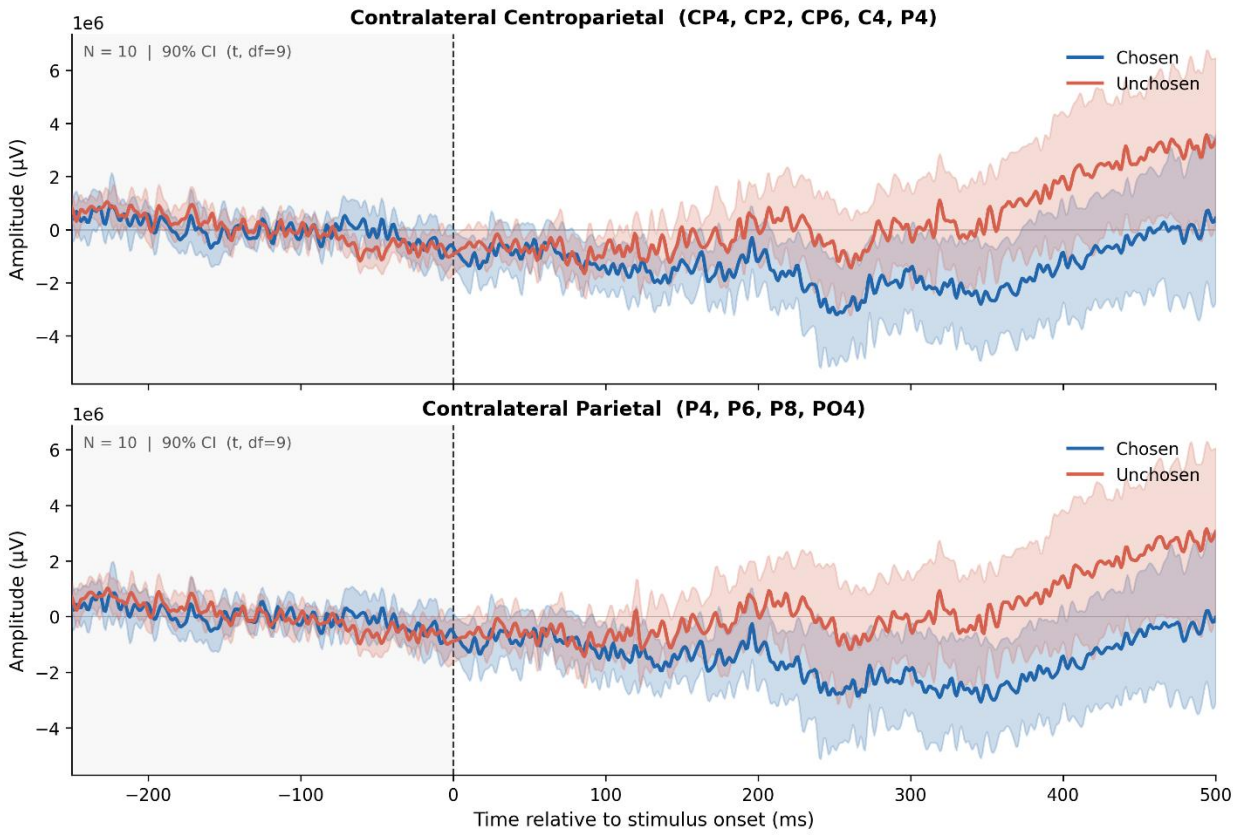


Figure 24: Chosen vs Unchosen ERPs for Centroparietal and Parietal ROIs. Across ROIs, a clear difference in neural amplitude between classes is shown after the P200 event, with the unchosen stimulus correlating with a higher neural amplitude. These results suggest that the chosen stimulus is typically associated with a lower overall intensity and area of sensation, matching the subjective reporting that typically found the triangular waveforms as preferred.

With the visual confirmation of an ERP difference, we sought to train a machine learning model that could classify epochs as either chosen or unchosen using features extracted from the ERPs. The specific machine learning model approach used was a linear discriminant analysis (LDA), chosen for its ability to reduce data dimensionality while maximizing class separation. Originally, two LDA models were trained individually on each of the two ROIs shown in figure 24. The model trained on the Contralateral Centroparietal ROI reached a 57.1% accuracy, whereas the model trained on the Contralateral Parietal ROI reached a 56.3% accuracy. Both of these models significantly differed from chance levels of 50% ($p = 0.002$ and $p = 0.003$, respectively).

Although these accuracies are significantly different from chance levels, the inclusion of multiple ROIs to a single classification algorithm could improve the classification accuracy. It is important to note that the inclusion of various classifiers to a single model results in harder to interpret results, as the final output depends on features from various ERPs that may not be directly related to each other. However, as an exploratory tool, this classification algorithm will prove if it is possible to classify chosen or unchosen epochs based purely on EEG data. Each epoch was limited from 50 ms pre-stimulation to 200 ms post-stimulation to ensure that the information fed into the LDA was directly correlated with the stimulation itself rather than other external events. Although this omits the P300 event, the robustness provided over the various classifiers offsets the possible reduction in information. From this base, 903 classifiers were extracted from the epoch data. The breakdown of these classifiers is as follows.

1. 21 unique electrode groups, each representing a spatial region of the scalp. Of these 21, 15 groups consisted of single electrodes or regional groups, whereas 6 consisted of bilateral pairs

2. 9 sliding time windows, each 50 ms with a 25 ms overlap, representing different ERP locations

3. 5 frequency bands, representing different frequencies of brain activities

For the alpha and broadband frequency bands, the -50 ms - 0 ms time window was omitted to reduce the effect of pre-stimulus noise on the overall training dataset. In total, this yields 21 groups and 43 band-window combinations, producing 903 base classifiers. For each classifier, 36

features were extracted from its corresponding electrode group's averaged EEG epoch within that classifier's defined time window and frequency band.

Importantly, the LDA operates on trial pairs, rather than individual trials. For each trial, a pair of one chosen epoch and one unchosen epoch is created. Each of these epochs contains the same number of unique features to be extracted. To highlight the differences between the chosen and unchosen features, the paired difference for each feature is calculated to indicate how much the two epochs differ. The sum of each feature is also calculated to represent the overall magnitude of activity for that trial. These sum-and-difference feature vectors are used directly to train the LDA model, rather than the features of an individual epoch within a pair. This is done to highlight the contrast between each trial's chosen and unchosen epochs. This paired construction is also used to serve as a data augmentation strategy. It follows that within one trial, if the first waveform was chosen over the second waveform, then the second waveform is not chosen over the first waveform. Thus, each trial pair can be represented by two distinct pairs of sum and difference feature vectors, rather than just one singular pair. For the chosen waveform, its difference vector was calculated as the chosen feature minus the unchosen feature. Conversely, the unchosen waveform's difference vector was calculated as the unchosen feature minus the chosen feature. The sum vectors of both the chosen and unchosen remain the same. This strategy doubles the number of training samples by using each epoch in a pair to produce individual training samples, yielding a larger training set than simply providing a single difference vector per pair.

For lateral electrode pairs, an additional spatial differencing step is applied before the paired augmentation. Before computing the differences between chosen and unchosen features, three representations of hemispheric asymmetry are computed for each epoch. These

representations are the raw, normalized, and absolute differences. These three representations highlight the absolute magnitude of asymmetry, the proportional dominance of one hemisphere over the other, and the degree of asymmetry, respectively. These representations are calculated for each feature, yielding a 108-element feature vector per epoch. Once this feature vector has been calculated for both the chosen and unchosen epochs, the difference and sum vectors are computed in the same way as for the non-lateral electrodes.

To eliminate the possibility of the model being trained on one subject's data and tested on that same subject, stratified group k-fold cross-validation was used. This method maintains that all of a subject's data is contained within a single fold. With ten participants, five folds of two participants each were used: four for training and one for testing. Further, each fold contains the same number of chosen and unchosen epochs to ensure that individual folds do not overfit to either class. The use of these folds ensures that the model's output showcases its generalizability to new subject data. The different combinations of subjects in these folds are referred to as “seeds”. Testing on different seeds ensures that the averaged model output is not heavily reliant on artifacts specific to certain subjects that would not generalize to unseen subjects.

Further, to reduce the effect of impedance mismatches between electrodes across participants (which affect the raw amplitude of the EEG data), subject-level Z-score normalization was applied to each subject's amplitude-dependent features. This ensures that amplitude variations are provided to the LDA on a comparable scale across all subjects, eliminating the possibility that one participant's high amplitude would dominate the features. For example, consider two subjects with individual peak amplitudes of 200 uV and 50 uV and average epoch amplitudes of 100 uV and 25 uV. Using subject-level Z-score normalization, each feature value is expressed as the

number of standard deviations from that subject's mean across all trials. Thus, both subjects' peak amplitudes would be normalized to a value of +2, preventing the LDA from assigning false importance to cross-subject amplitude differences.

Once the LDA base classifiers were trained, an L1-regularized logistic regression meta-learner was fit to the obtained out-of-fold (OOF) probability matrix. Each column of this matrix contains the held-out probability predictions from one base LDA classifier across all training samples. The meta-learner trains on data that the base classifiers never saw during their own training, preventing leakage between the two model levels. This step induces sparsity, as classifiers with redundant values are assigned a weight of zero, thereby making the overall algorithm easier to interpret. The meta learner then sweeps through various regularization strength coefficients for each chosen seed to maximize the model's balanced accuracy (BA). The regularization strength coefficient typically varied between seeds, showcasing that the appropriate degree of sparsity depends on the subject-specific test data. After this sweep, the model with the highest balanced accuracy was chosen to ensure it achieved similar accuracies across classes.

The result was validated against a null distribution using a label-shifted permutation test (N = 50 iterations), in which the full model pipeline was re-run on randomly permuted class labels. The permuted model achieved a mean balanced accuracy of 49.9% (maximum across all 50 runs: 53.6%), confirming that the true model's results reflect a genuine learned signal rather than any statistical artifact of the data structure ($p < 0.02$). The final output of the cross-validated pipeline substantially exceeds the null distribution, yielding an average balanced accuracy of 77.3% (±1.9%) across six fixed cross-validation seeds [0, 1, 7, 13, 42, 100], with a mean AUC of 0.847. The confusion matrices for these seeds are shown below, reported with their optimized

regularization coefficient.

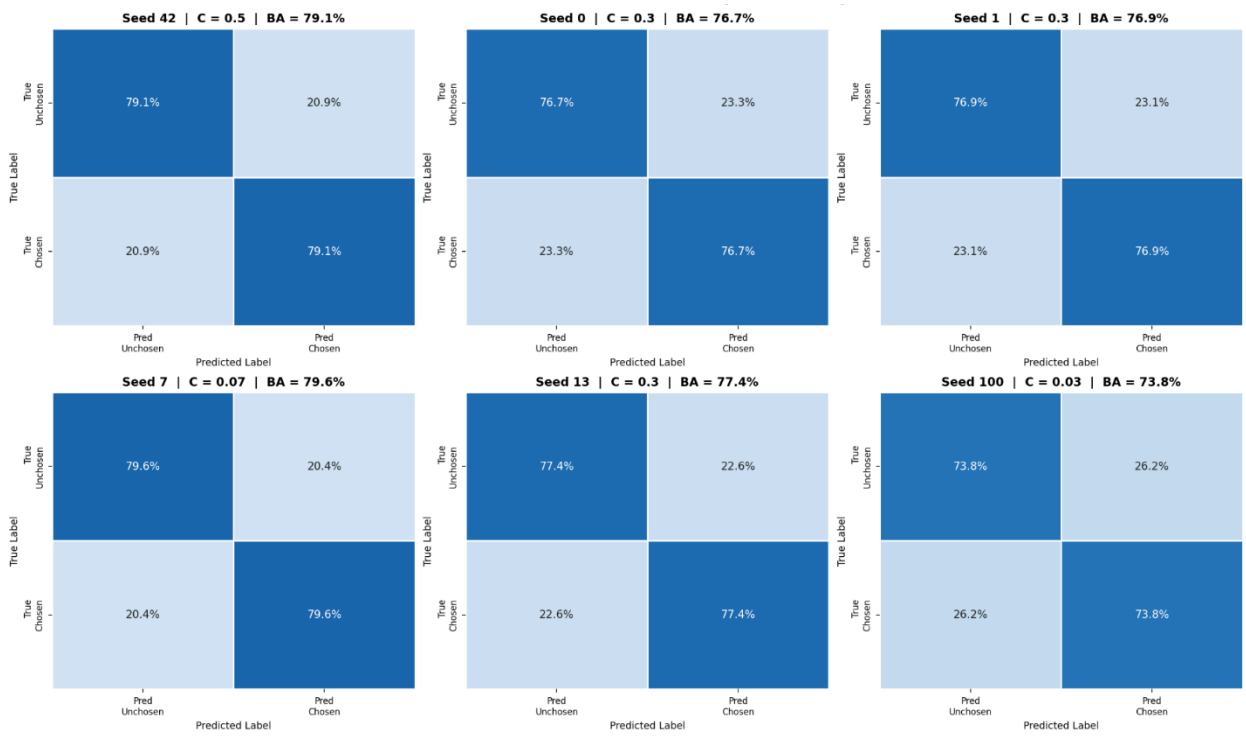


Figure 25: Per-Seed Confusion Matrices. Each seed chosen displays significantly higher accuracy than the chance level of 50%

When the predicted probabilities are averaged across all six seeds prior to thresholding, the consensus balanced accuracy rises to 83.6% with an AUC of 0.909, reflecting a reduction in seed-

specific noise from combining model outputs. This can be seen in the confusion matrix below.

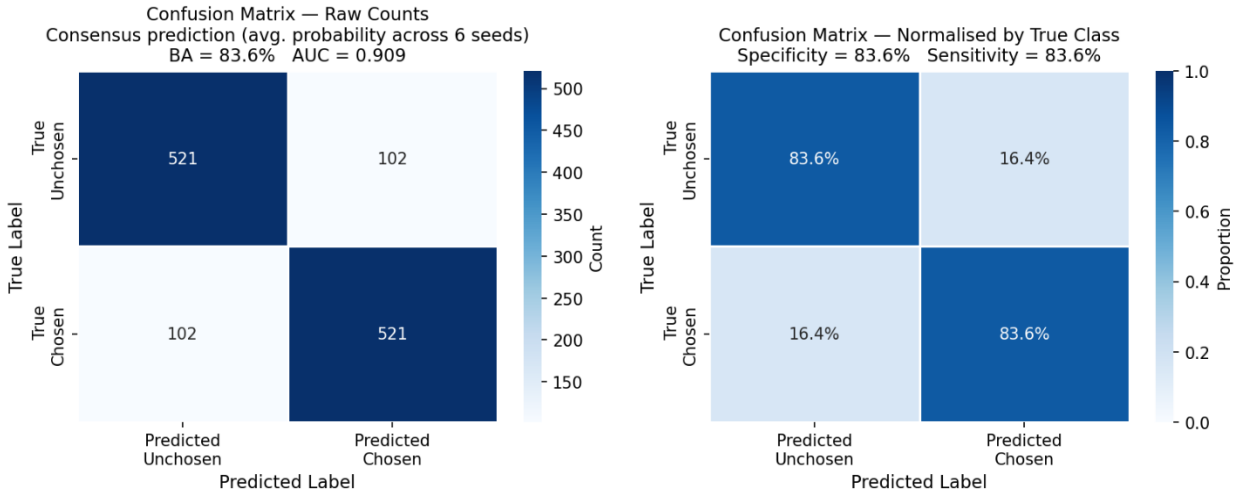


Figure 26: Optimal classification through averaging probabilities across seeds. Although this approach achieves a higher accuracy at 83.6%, the accuracy associated with the model should be related to the per-seed averages.

Notably, the alpha band, which is linked to somatosensory activities, was found to be the fourth most important frequency band for this classifier. This is likely due to the alpha band consisting of a small spectral frequency, and thus contains a limited amount of "useful" data compared to larger bands. This hypothesis is strengthened, as the beta band was the most impactful, with 71% of its classifiers selected for the final model, followed by the gamma bands. The theta band was tied with the alpha band for percentages of classifiers chosen. Broadband EEG features contributed the least to the classifier performance, with only 58% of their classifiers retaining non-zero weights. Although the broadband has the widest frequency band, its applicability was likely reduced due to the significant amount of interfering information it presents, likely canceling out

significant features that were highlighted in specific bands.

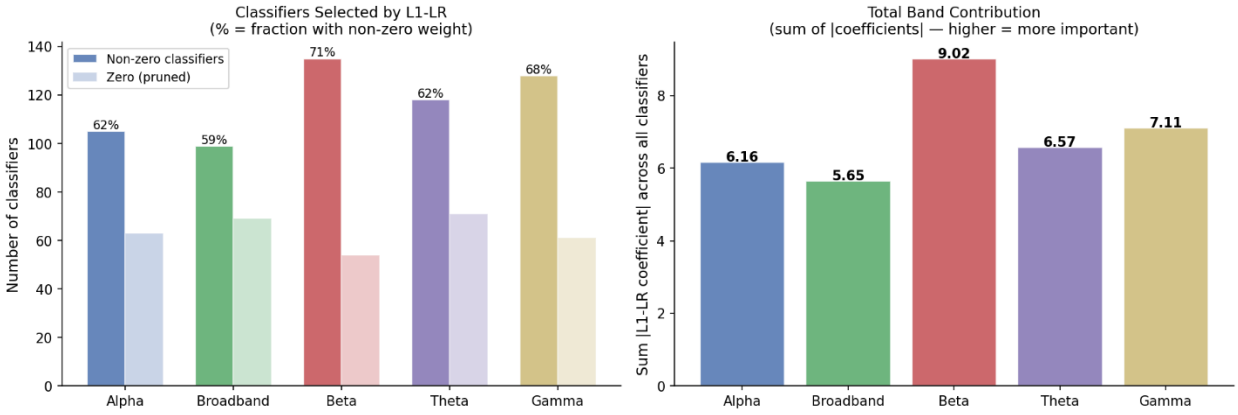


Figure 27: Per-Band Classifier Contribution. The broadband contributed the least to the final model, likely due to the large interference between important SEPs that occurred throughout its wide frequency range. Beta and gamma bands were found to contribute the highest, followed by the theta and alpha bands.

Specific feature importance highlighted the significance of the C, CP, and FC electrodes, as well as the connections formed by lateral pairs of these electrodes (e.g., C1-C2, C3-C4). A detailed breakdown of the 25 base classifiers that the meta learner assigned the largest mean L1-Regularised coefficients to, averaged across the six fixed cross-validation seeds, is shown below.

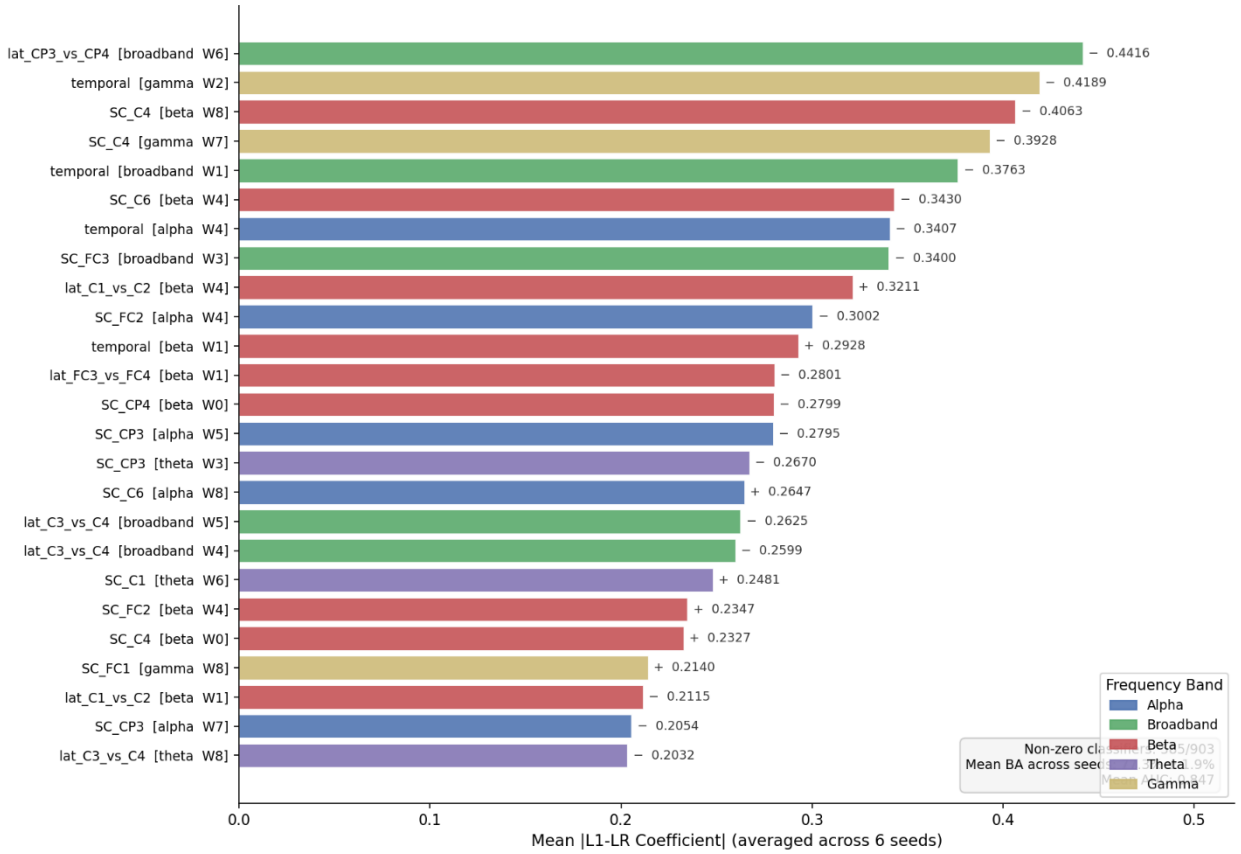


Figure 28: Top 25 Base Classifiers by Weight. Negative weights represent classifiers that had their probabilities flipped to produce a higher accuracy. Interestingly, the classifier with the highest weight was found within the broadband frequency band.

The most important classifier with a positive weight is shown below, with the specific window that features were extracted from highlighted in yellow. A clear visual difference between the chosen and unchosen epochs is evident. This clear difference demonstrates why this classifier was assigned the highest weight, as the extracted features likely differed greatly.

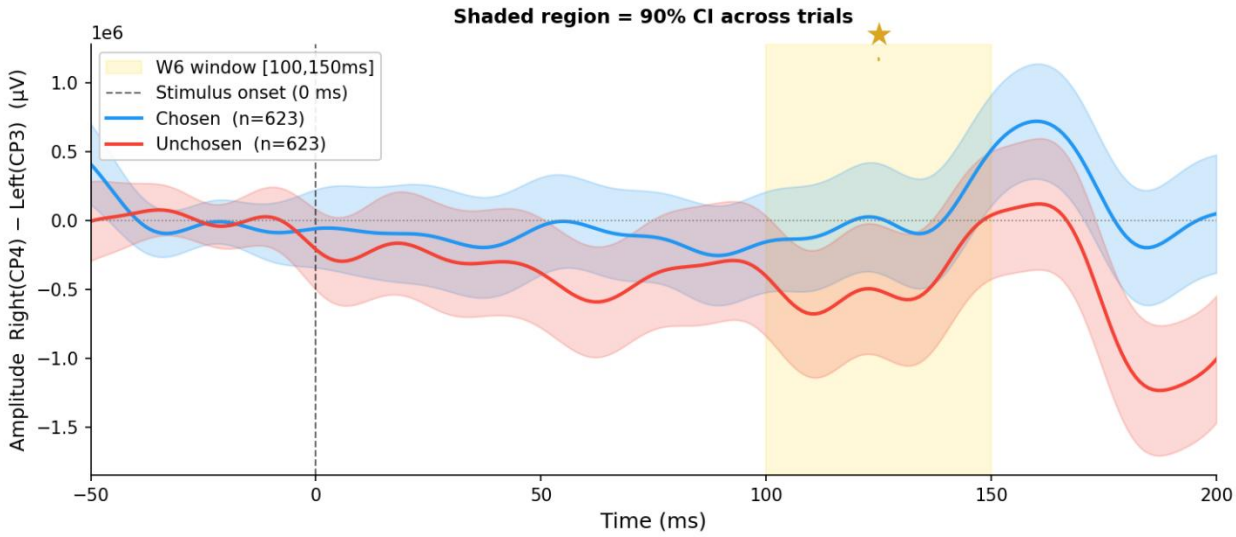


Figure 29: Single most important classifier with a positive weight. Within this band, it shows clear separation between chosen and unchosen stimuli, with a relatively compact 90% confidence interval.

DISCUSSION

This thesis is a deep investigation into the subjective and objective effects of five temporally unique stimulus waveforms during TENS of the median nerve. The goal of this thesis was to determine and highlight differences between the qualities of the evoked sensation and relate these differences to their driving factors, as well as provide insight into the neural representations that may explain these subjective differences. Subjective data consisted of self-reported information concerning the area in which an evoked sensation occurred, as well as its specific sensation type, and a two-alternative forced-choice task. Objective data consisted of a 64-channel EEG headcap, which produced relevant SEPs concerning the evoked sensations across 5 unique ROIs.

Subjective data analysis revealed relationships between charge, waveform, evoked sensation intensity, and overall area. Waveform shape influenced the amount of charge required for a sensation to be evoked, as each waveform differed in threshold charge. On average, the square waveform had the highest overall threshold charge, followed closely by sine. The three triangular waveforms each required a lower charge to evoke a sensation, suggesting that these waveforms might produce "softer" sensations. The relationship between charge and perceived intensity is almost one-to-one, with only the linear increase waveform swapping positions with the linear decrease waveform between the two categories. Interestingly, at the base amplitude of stimulation, the sine waveform has a higher perceived intensity than the square waveform, even with a smaller charge. At the mid and high amplitudes, the square waveform is perceived as more intense, likely due to the charge difference between the two waveforms. This suggests that the shape of a waveform can be the driving factor behind perceived intensity; however, its influence greatly diminishes as the charge mismatch between comparisons increases.

The square and the sine waveforms were also found to evoke sensation over a larger overall area of the hand than the three triangular waveforms, although these results are not found to be statistically significant due to high inter-subject variation. The sine waveform evoked a larger overall area of sensation than the square waveform at each of the three amplitudes of stimulation, even with a smaller charge. As the charge difference increased, its influence on area of sensation became apparent, as at base amplitude the difference in area between sine and square was 1.1%, whereas the difference at high amplitude is 0.5%. This suggests that the waveform shape has an influence on evoked sensation area that is more resistant to charge mismatches than its relationship with intensity. The underlying mechanism behind both of these observed relationships suggests that shape and charge exhibit a positive relationship between both perception qualities. Unique

waveforms resulted in unique areas of sensation across the hand, although all five waveforms were able to evoke sensation across the palm. The sine waveform was able to consistently evoke sensation on the participant's thumb, and both the sine and square waveforms were able to consistently evoke sensation on the participant's middle finger.

Unique sensations were evoked across the five waveforms. The square waveform correlated with a significantly higher electric sensation, and the sine waveform correlated with a significantly higher pulse sensation. Interestingly, no waveform was able to evoke a significant intensity of the touch sensation. These sensations were found to be evoked with different average charges. The pulse and electric sensations averaged 0.06 μC more charge than the other six sensations, which were found to be experienced on average within a 0.02 μC range. The relationship between charge and perceived intensity was not present, highlighting the importance of waveform shape. Specific sensations were evoked at a higher rate between waveforms, such as pulse for sine (53%) and touch for linear decrease (15%). Similarly, specific sensations were evoked at a lower rate between waveforms as well. The rate of recruitment was found to be charge-dependent, as the touch sensation was recruited at a higher rate at the base and mid charges than the high charge for all waveforms besides linear increase. This relationship is waveform-specific and provides a method for optimizing recruitment rates for desired sensations.

These sensations were experienced on different areas of the hand, although their areas were found not to share a direct relationship to their associated charges. This is likely due to the sensations being averaged over the five different waveforms, as well as the high inter-subject variation in subjective experience. The pulse sensation consistently evoked sensation on the middle finger of the participant, while the needle sensation consistently evoked sensation on the thumb.

The tingling and electric sensations were able to evoke sensation on all fingers that corresponded to the median nerve.

Objective data analysis was first used to compare SEPs between the five waveforms within the two uniquely defined ROIs. Within the centroparietal ROI, the mean amplitudes between the five waveforms significantly differed around the N140 ($p = 0.005$) and P300 ($p = 0.004$) events. The amplitude ordering around these events (sine > square > triangular) almost directly corresponds to the order in which waveforms were able to evoke the largest area of sensation. The parietal ROI revealed significantly different mean amplitudes between waveforms near the P200 ($p = 0.035$) and P300 ($p = 0.01$) events. Interestingly, near the P200 event, the waveform with the highest amplitude switches from sine to square. The order observed in the Centro-parietal ROI is restored around the P300 event. The significant events provided by these SEPs suggest that the relationship between perceived sensation and neural activity is multifaceted, depending on both the area in which the evoked sensation occurs and its intensity. Further, the largest magnitude of influence that each of these sensation categories has on neural activity could be temporally separated, as seen within the SEP generated by the parietal ROI. Based on this suggestion, it can be inferred from the literature that the intensity of an evoked sensation correlates with a higher neural amplitude during the perceptive stage of sensory processing. It should be noted that the general findings presented across SEPs could be driven purely by a combination of the intensity and area of evoked sensation, as the sine waveform, which typically had the highest amplitude, also had the highest sensory drive score out of the five waveforms.

SEPs within the parietal ROI were then compared within waveform class between their three possible amplitudes: base, mid, and high. The square waveform showcased clear differences in neural amplitudes between the three stimulus amplitudes around the P200 and P300 events.

Interestingly, the base stimulus amplitude correlated to a higher neural activity than the mid amplitude around the P200 event, even though it correlated with a lower overall intensity and area of sensation. The high stimulus amplitude correlated to the highest amplitude of neural activity across the majority of the SEP. The sine waveform similarly exhibited notable visual differences in neural amplitudes between the three amplitudes of stimulation around the same two events. For the sine wave, the expected order of high, mid, and low matched the neural amplitudes, unlike the square wave. All three amplitudes of the sine wave correlated to a higher neural amplitude than the three amplitudes of the square wave, respectively, around the P300 event. This higher neural amplitude could result from the sine wave evoking a larger area of sensation across the hand for each amplitude of stimulus. The three triangular waveforms did not result in any visual differences between their three stimulus amplitudes, highlighting the limited effectiveness of EEG data when dealing with stimuli that result in low intensity and areas of evoked sensations.

Comparison between SEPs generated by ROIs based on Alibou et al. and the SEPs presented within that paper was limited. The central frontal and contralateral parietal ROIs did not showcase any similarities between expected significant events within their SEPs. The only comparable event present between studies was found in the superior parietal ROI, which produced a similar visual P300 peak across all five waveforms. This peak occurred much later within the P300 window than expected. This general lack of similarities in events showcases the large inter-subject variation that EEG data presents, as many of these expected events may have averaged out over the eleven participants whose data were utilized for this analysis. The ERSPs of these same ROIs showcased significant synchronization within the theta and alpha bands across all five waveforms within all three ROIs, in agreement with Alibou et al. However, these ERS's were

limited in spectral range in comparison with the limited ERSs that concerned the beta band. Those that did were much less significant than the ERS's in the lower frequency bands. The most significant ERS's within the high beta band occurred for the linear increase and decrease waveforms within the central frontal ROI. ERD varied in occurrence between waveforms, occurring in the beta band for the central frontal ROI for the linear increase and decrease waveforms, the superior parietal ROI for the sine waveform, the contralateral parietal ROI for the square waveform, and across all ROIs for the triangular waveform.

The two-alternative forced-choice task revealed a clear ability across participants to correctly distinguish an individual waveform from the others, even while blinded. Participants consistently identified a waveform within a set of two with an 83.5% accuracy, significantly higher than the chance level of 68.29% ($p < 0.001$). The waveform identified as the most preferred varied widely across subjects, with the most preferred waveform, linear decrease, being selected by only 40% of participants. This highlights the large variation of subjective experience that occurs even across similar stimulation parameters (the amplitude of stimulation varies based on the per-subject threshold). Interestingly, this finding matches the subjective reporting presented by Collu et al., as the linear decrease waveform displayed the highest natural sensation value within that study. Within this study, the linear decrease waveform did not display the highest natural sensation value but was selected as the preferred waveform by the most participants during this task, highlighting the large variation present subjective self-reported data. The square waveform was selected as the most preferred waveform by 10% of participants, suggesting that the traditional standard waveform for stimulation is almost always a suboptimal waveform, if the goal of the stimulation is to evoke a "natural" sensation.

SEPs were analyzed within the two user-defined ROIs to determine if a difference in neural activity could be seen between the stimulation trials that were chosen by the participants as preferred, and those that were not. Within both the Centro-parietal and parietal ROIs, a clear difference between the classes could be seen around the P300 event, with the unchosen trials displaying a higher amplitude of neural activity than the chosen trials. This suggests that the chosen trials, on average, were perceived as less intense than the unchosen trials [33]. This suggestion closely matches the conclusions of the two alternative forced-choice tasks, as the linear decrease waveform that was selected as preferred by the most participants evoked the 2nd least intense sensation over the 3rd smallest area, while the square waveform that was selected the least by participants evoked the 1st most intense sensation over the 2nd largest area. Broadly, these findings suggest that sensations that are perceived as preferred strongly correlate with sensations that are not intense, and could correlate with sensations that are relatively targeted to a small area.

Introductory attempts to use a machine learning classification algorithm to classify the unchosen and chosen epochs exhibited a limited success rate when trained on data from SEPs over individual ROIs. The LDA model trained on the Centro-parietal ROI reached 57.1% accuracy, whereas the model trained on the parietal ROI reached 56.3%. Although these are both significantly higher than the chance level of 50% ($p = 0.002$ and $p = 0.003$, respectively), these accuracy levels could be significantly improved through the inclusion of information from SEPs across multiple ROIs. A preliminary LDA machine learning approach was investigated, using 903 base classifiers across 21 unique ROIs. The inclusion of these various classifiers resulted in a final accuracy for the highest seed being 79.6%. Although this result presents limited application due

to the complexity of training, it showcases that information contained within EEG data can be used with relatively high accuracy to determine if an individual prefers a certain stimulus, even when trained on data across participants.

The observed relationships highlighted through subjective data provide insight into possible optimal waveforms for certain scenarios. The triangular waveforms were consistently able to evoke sensation with a lower required charge, were found to be preferred by 70% of participants, and correlated with the highest rate of recruitment for the touch sensation. Thus, these waveforms should be used when the goal of a study is to evoke close to natural sensations. Depending on the specific triangular waveform used, the overall charge above threshold can be determined to optimize the rate of recruitment. The sine waveform was shown to evoke sensation over a wide area of the hand, and thus would be best used in haptic feedback studies that have less concern over sensation quality. The square waveform was able to evoke the most intense sensation on average, and thus could be used similarly to sine for haptic feedback applications that indicate time-sensitive events. Unfortunately, conclusions between driving factors and specific areas of sensation were limited. Broadly, shape and charge were found to have a positive relationship with both area and intensity (e.g., if a shape/charge increases area, it will also increase intensity, and vice versa)

Various limitations were present within this study. The overall participant base is relatively small (n = 11), with a clear gender gap (9 males, 2 females). Due to the sheer number of comparisons in this data set, it is unlikely that some statistical significances would remain after correction. Further, EEG and ERSP data analysis was mainly qualitative, with few statistically significant events. The distortion of waveforms likely experienced during stimulation may reduce the accuracy of the final conclusions about the relationship between shape and experience, as

participants in future studies may have different physical attributes that distort these waveforms in different ways.

CONCLUSION

This study presents a diverse dataset concerning participants' subjective experience and objective brain data (EEG) during TENS utilizing five unique stimulation waveforms. These unique waveforms were able to evoke significantly different perceived intensities of evoked sensation, in notably different areas of the hand, across all participants. As expected, the increase in charge resulted in an increase in perceived intensity and area of the evoked sensation regardless of the stimulus waveform. Broadly, shape and charge have complementary effects on intensity and area, positively or negatively influencing both in tandem. Stimulus waveforms produced unique, charge-dependent rates of recruitment for specific sensations across participants, highlighting the influence shape has on sensation. A two-alternative forced-choice task found that 90% of participants preferred non-rectangular waveforms, suggesting that the traditional square waveform used for TENS typically results in suboptimal sensations. EEG data suggest that the perceived intensity and area of a sensation jointly influence the amplitude of neural activity. These two categories may correspond to unique SEPs, specifically intensity to N140 - P200 and area to P300; these conclusions require further analysis. Inter-subject variability within EEG data proved to eliminate various SEPs that were present in a similar study of a single participant. Among stimulus waveforms that evoked high perceived intensities, EEG data clearly showed separation around the P300 event between different overall charges of stimuli within waveform class, for the sine and square waveforms. EEG data further showcased clear separation between chosen and unchosen stimuli across waveforms during the two alternative forced choice tasks around the P300 event. In

its entirety, this study draws introductory conclusions concerning the relationships between stimulus waveforms, perceived sensations, and physiological responses. Subjective reporting suggests that triangular waveforms should be used in studies where evoked "natural" sensations are desired, due to the relatively high rate of recruitment for the touch sensation and its low rate of recruitment for the pulse and electric sensations. The sine and square waveforms are better suited for studies that desire broad areas of noticeably intense sensation, which are less concerned with the type of sensation itself. Further research investigating how to increase control over the exact locations and sensations evoked by these stimulus waveforms must be performed to increase their impact on neuroprosthesis.

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